

**Assessing the Environmental Cost of
High-Performance Computing**
*From Operational Scheduling to Lifecycle
Sufficiency*

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This is the original version of the
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to the Examining Committee.

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Resumo

Lucas de Sousa Rosa. **Avaliando o Custo Ambiental da Computação de Alto Desempenho: Do Escalonamento Operacional à Suficiência no Ciclo de Vida**. Exame de Qualificação (Doutorado). Instituto de Matemática, Estatística e Ciência da Computação, Universidade de São Paulo, São Paulo, 2026.

A computação de alto desempenho impõe custos ambientais em carbono, água e materiais ao longo de todo o seu ciclo de vida, mas a área tradicionalmente equipara esses custos ao consumo operacional de energia. Este trabalho procura entender o que a área mede, o que ela deixa de fora e como sistemas de alto desempenho podem ser operados e avaliados quando as dimensões menos estudadas são incorporadas. Ele começa com um estudo de mapeamento sistemático de 62 relatos que cobrem dezesseis anos de pesquisa, classificados segundo impactos ambientais, estágios do ciclo de vida, locais do sistema e mecanismos de gestão e intervenção. A literatura se concentra em um único corredor dessa classificação: carbono, estudado no estágio operacional, no nível da infraestrutura e por meio de medição. Carbono aparece em 54 dos 62 relatos, materiais e resíduos em 12, uso de água em 5 e biodiversidade em 1, e nenhum relato trata mais de dois impactos ao mesmo tempo. O corredor acompanha a disponibilidade de dados mais do que o formato do problema, e são muitas as células que ele deixa vazias. O restante do trabalho segue duas delas. A primeira permanece na operação, onde escalonadores, plataformas e decisões de substituição de hardware raramente precificam o custo ambiental. Estendemos o SimGrid e o Batsim com um modelo de pegada por host que contabiliza carbono operacional, água local e remota e impactos incorporados sob sinais de rede elétrica variáveis no tempo, e o usamos para avaliar o Greenfilling, uma regra que suprime inícios por backfilling quando a intensidade da rede excede sua média móvel. Em 288 comparações pareadas sobre três redes nacionais e duas máquinas de produção, nenhuma mediana de cenário alterou a pegada alvo em 0,1%, enquanto o slowdown limitado médio aumentou em todos os pares, de modo que uma política precisa precificar tanto a espera quanto a ação, e não apenas recusar. Uma análise offline das mesmas cargas de trabalho situa o teto do deslocamento de tarefas em 6,76% do carbono operacional e 4,89% da água, com deslocamento máximo de 24 horas. A segunda acompanha como os sistemas são julgados. O Green500 os ordena por desempenho por watt, uma razão que pode melhorar enquanto o impacto absoluto por trás dela cresce. O plano desenvolve mecanismos que precificam o custo ambiental no escalonador, nos estados de energia da plataforma, no momento da substituição de hardware e junto aos usuários, e uma avaliação orientada à suficiência que usa o Green500 como ponto de referência para expor o que uma razão por watt esconde.

Palavras-chave: Computação de alto desempenho. Impacto ambiental. Sustentabilidade. Pegada de carbono. Pegada hídrica. Escalonamento de tarefas. Avaliação de ciclo de vida. Suficiência.

Abstract

Lucas de Sousa Rosa. **Assessing the Environmental Cost of High-Performance Computing: From Operational Scheduling to Lifecycle Sufficiency**. Qualifying Exam (Doctorate). Institute of Mathematics, Statistics, and Computer Science, University of São Paulo, São Paulo, 2026.

High-performance computing carries environmental costs in carbon, water, and materials across its whole lifecycle, yet the field has traditionally equated those costs with operational energy consumption. This work tries to understand what the field measures, what it leaves out, and how HPC systems can be operated and evaluated once the less studied dimensions are brought in. It begins with a systematic mapping study of 62 reports covering sixteen years of research, classified along environmental impacts, lifecycle stages, system loci, and management and intervention levers. The literature concentrates in a single corridor of that classification: carbon, studied at the operational stage, at the facility level, and through measurement. Carbon appears in 54 of the 62 reports, materials and waste in 12, water use in 5, and biodiversity in 1, and no report addresses more than two impacts at once. The corridor traces the availability of data more than the shape of the problem, and the cells it leaves empty are many. The rest of the work follows two of them. The first stays with operations, where schedulers, platforms, and hardware-replacement decisions rarely price environmental cost. We extended SimGrid and Batsim with a per-host footprint model that accounts for operational carbon, on-site and off-site water, and embodied impacts under time-varying grid signals, and used it to evaluate Greenfilling, a rule that suppresses backfill starts when grid intensity exceeds its moving average. Across 288 paired comparisons over three national grids and two production machines, no scenario median moved the targeted footprint by 0.1%, while mean bounded slowdown rose in every pair, so a policy has to price both waiting and acting rather than only refusing. An offline analysis of the same workloads places the ceiling of job shifting at 6.76% of operational carbon and 4.89% of water, at a maximum displacement of 24 hours. The second follows how systems are judged. The Green500 ranks them by performance per watt, a ratio that can improve while the absolute burden behind it grows. The plan develops mechanisms that price environmental cost in the scheduler, in the platform's power states, in replacement timing, and with the users, and a sufficiency-aware evaluation that uses the Green500 as a reference point to expose what a per-watt ratio hides.

Keywords: High-performance computing. Environmental impact. Sustainability. Carbon footprint. Water footprint. Job scheduling. Lifecycle assessment. Sufficiency.

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Chapter 1

Introduction

High-performance computing (HPC) has become essential to modern science, and its capability has grown by drawing more power, packing more hardware, and cycling through that hardware ever faster (PORTEGIES ZWART, 2020; ALLEN, 2022). The environmental cost of this growth has traditionally been equated with operational energy consumption (KOOMEY *et al.*, 2011), yet operational energy is only one facet of the footprint. Manufacturing the processors, memory, and interconnects that populate a machine emits carbon long before the system is switched on, and this embodied share is rising as fabrication grows more complex (GUPTA *et al.*, 2021; PIRSON *et al.*, 2023). Cooling those systems and generating the electricity that feeds them consumes freshwater at scales large enough to matter regionally (MYTTON, 2021). The semiconductor supply chain draws on scarce materials and discharges hazardous waste (SANDHU *et al.*, 2025), while the shortening replacement cycles that keep systems competitive turn working hardware into electronic waste. Across its full lifecycle, then, an HPC system carries costs in carbon, water, and materials and leaves electronic waste behind, well beyond the operational energy it draws.

Visibility into this broader footprint has only recently begun to grow, and the established ways of evaluating and operating HPC systems still capture only part of it. The Green500 ranks systems by performance per watt under a single benchmark (FENG and CAMERON, 2007), reducing sustainability to an efficiency ratio measured on Linpack. Job schedulers decide when and where computation runs, yet they have conventionally been designed to prioritize performance above all else (KOCOT *et al.*, 2023; CZARNUL *et al.*, 2019), leaving environmental cost outside the optimization objective. Procurement and hardware-replacement decisions weigh performance and acquisition cost while treating lifecycle impact as an afterthought (LI *et al.*, 2023). The research concerning these overlooked aspects has started to gain traction only lately. At the same time, current strategies target just one piece of a challenge that is not yet fully understood. Even the most closely monitored indicator shows no net progress, because rising demand continually cancels out any benefits achieved (YORK *et al.*, 2022).

To translate this rising attention into well-defined research needs, the project begins with a systematic mapping study (ROSA *et al.*, 2026). Classifying 62 reports from sixteen years of research across environmental impacts, lifecycle stages, system loci, and

management levers, the study finds the literature concentrated within a narrow path. In particular, existing papers focus mostly on carbon among the impacts, operations among the lifecycle stages, facility infrastructure among the system loci, and measurement among the management levers. Meanwhile, other topics appear only sporadically and are analyzed in isolation, such as non-cooling water usage, biodiversity, and end-of-life processing. To address these gaps, the rest of this work divides into two branches (Figure 1.1).

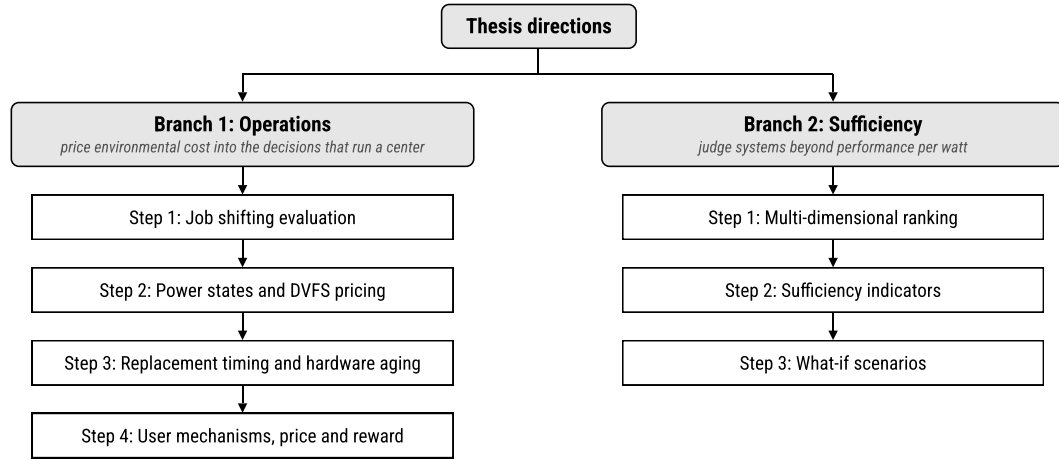


Figure 1.1: The two branches of this work and the steps of each. Steps are ordered by dependency.

The first branch focuses on the operation of HPC systems. To evaluate different management strategies without needing access to a production system, we built a simulation testbed. The testbed has so far introduced footprint models into existing simulators to enable the evaluation of operational carbon, operational water, and embodied footprint. A first result along this line is Greenfilling, a small change to EASY backfilling (SKOVIRA *et al.*, 1996) that holds back opportunistic job starts when the targeted grid intensity, carbon or water, runs above its recent average. Evaluated across three national grids and two production machines, it traded certain delay for uncertain environmental return. This result sets a lower bound on how sophisticated such an intervention has to be. A job-shifting variant was also developed. It prices delaying each job against the grid intensities it would meet, and an offline analysis of the same workloads and signals places its ceiling near 7% of operational carbon, two orders of magnitude above what Greenfilling reached. We plan to extend this testbed by including user-facing accounting, platform controls, and lifecycle-aware policies.

The second branch focuses on the overall impacts of HPC systems. The Green500 ranks systems by performance per watt, but this is only one facet of the broader impacts of using those platforms. Two systems can rank closely in the Green500 while drawing very different absolute power, sitting on grids whose water and carbon intensity differ by orders of magnitude, carrying different embodied impacts, and consuming water at rates that do not reflect regional scarcity. By applying a sufficiency perspective to Green500 data, we intend to examine cases where efficiency improves while absolute burden grows, where embodied costs dominate but stay invisible, and where regional context changes which system is actually less harmful.

The remainder of this document is organized as follows. [Chapter 2](#) reviews the environmental footprint of HPC across its lifecycle, then presents the systematic mapping study and the classification behind the gap this work addresses. [Chapter 3](#) reports the preliminary results. [Chapter 4](#) sets out what remains. It adds user modeling, platform controls, and lifecycle-aware policies to the testbed built in the first branch, and designs a sufficiency-aware evaluation that uses the Green500 as a reference point, widening the ranking to the impacts efficiency omits and reaching toward the absolute burden a per-watt ratio hides, as far as public data and what-if scenarios permit. [Chapter 5](#) closes with the schedule and the expected contributions of the completed work.

Chapter 2

The Environmental Impacts of HPC

Before tackling the problem, we need to understand how the existing literature sees it. A systematic mapping study (SMS) scans a body of work to answer what has been studied, how it has been studied, and where the literature runs thin. This chapter presents the resulting map of the environmental impacts of high-performance computing. The SMS was conducted following the guidelines of Petersen et al. (PETERSEN *et al.*, 2008).

2.1 What the Existing Reviews Miss

Which environmental impacts does the field study, how does it address them, and what does it leave uncovered? These are the questions we set out to answer, and the existing reviews do not answer them.

Most of the existing work looks at operational energy and stays close to the running machine. O'Brien et al. (O'BRIEN *et al.*, 2018) survey predictive power and energy models for heterogeneous architectures, Czarnul et al. (CZARNUL *et al.*, 2019) catalog energy-aware tools such as DVFS and power capping, Kocot et al. (KOCOT *et al.*, 2023) review energy-aware scheduling, and Cai et al. (CAI *et al.*, 2011) organize the field around the trade-off between power and performance. Silva et al. (SILVA *et al.*, 2024) reach further and cover carbon and renewable management across infrastructure, hardware, software, and policy, yet the analysis still centers on the relation between operational energy and carbon, and leaves the wider environmental picture to future work. These reviews map operational energy well, but they do not set out to cover several environmental impacts or to follow a system across its lifecycle.

The broader sustainability literature on data centers repeats the pattern. Surveys there cover renewable integration and waste heat (SHUJA *et al.*, 2016), cooling systems (ZHANG *et al.*, 2021), thermal-aware scheduling (CHAUDHRY *et al.*, 2015), and operational metrics such as PUE and WUE (NAGARATHINAM *et al.*, 2024), and the most recent ones add explicit review protocols and richer operational measures (KAHIL *et al.*, 2025; SAFARI *et al.*, 2025; KUMAR and PINDORIYA, 2026). The scope stays on operational efficiency, and the setting is the data center rather than HPC.

A few reviews go beyond that scope. Whitehead et al. (WHITEHEAD *et al.*, 2014; WHITEHEAD *et al.*, 2015) show what a wider data-center assessment can cover, including energy, carbon, water, materials, human health, and waste disposal across the lifecycle, and D’Orgeval et al. (D’ORGEVAL *et al.*, 2024) propose a holistic framework that joins environmental, economic, and ecosystem concerns. These works show how embodied and operational impacts interact, and why operational energy alone is a thin basis for environmental judgment. They also step outside HPC to do it.

No review we found combines HPC scope, a systematic protocol, several environmental impacts, and lifecycle coverage (Table 2.1). The broad, lifecycle-aware reviews tend to look past HPC, and the HPC-focused ones tend to stay close to operational energy, so relatively little work sits at their intersection. That gap motivates a systematic mapping study, which applies an explicit protocol to a question the existing reviews mostly leave open, namely how the environmental impacts of HPC are discussed across the lifecycle.

Reference	Year	D1	D2	D3	D4	D5	D6	D7	D8
CAI <i>et al.</i>	2011		✓					✓	
WHITEHEAD <i>et al.</i>	2014			✓	✓	✓	✓	✓	✓
WHITEHEAD <i>et al.</i>	2015			✓	✓	✓	✓	✓	✓
CHAUDHRY <i>et al.</i>	2015			✓				✓	
SHUJA <i>et al.</i>	2016			✓				✓	
O’BRIEN <i>et al.</i>	2018		✓					✓	
CZARNUL <i>et al.</i>	2019		✓					✓	
ZHANG <i>et al.</i>	2021							✓	
KOCOT <i>et al.</i>	2023		✓					✓	
SILVA <i>et al.</i>	2024		✓	✓	✓	✓	✓	✓	✓
D’ORGEVAL <i>et al.</i>	2024			✓	✓	✓	✓	✓	✓
NAGARATHINAM <i>et al.</i>	2024			✓	✓	✓	✓	✓	
KAHIL <i>et al.</i>	2025	✓		✓	✓			✓	
SAFARI <i>et al.</i>	2025	✓		✓	✓			✓	
KUMAR and PINDORIYA	2026							✓	
Our study	2026	✓	✓	✓	✓	✓	✓	✓	✓

D1 systematic protocol, D2 HPC scope, D3 carbon emissions, D4 water use, D5 materials and waste, D6 production stage, D7 operational stage, D8 end of life.

Table 2.1: Related reviews and surveys compared along four dimensions: systematic methodology (D1), HPC scope (D2), breadth of environmental impacts (D3–D5), and lifecycle coverage (D6–D8). A review counts under D1 only if it documents a protocol with explicit inclusion and exclusion criteria and a screening process, following established guidelines such as those of Kitchenham et al. (KITCHENHAM and CHARTERS, 2007) and Petersen et al. (PETERSEN *et al.*, 2008).

2.2 How the Map Was Drawn

The corpus was drawn from three bibliographic databases: IEEE Xplore, ACM Digital Library, and Scopus. The search query followed a Population, Intervention, Comparison, and Outcome rationale (Table 2.2), pairing terms for HPC systems and infrastructure with terms for environmental assessment and sustainability, and it ran over titles, abstracts, and keywords. Inclusion and exclusion criteria were fixed before any record was read (Table 2.3). The three databases returned 4812 records, and 3788 remained once duplicates were removed.

Element	Definition	Terms used
Population	HPC systems and infrastructure	<i>high-performance computing, supercomputing, supercomputer, hpc</i>
Intervention	Environmental assessment or sustainability practices	<i>sustainability, sustainable, ecological, footprint, environmental impact, carbon emission, greenhouse gas, water consumption, water usage, lifecycle assessment, lca, embodied carbon, e-waste, electronic waste, material depletion, resource consumption, rare earth</i>

Table 2.2: PICO elements used to structure the SMS search string. Comparison and Outcome were not included, as they do not fit the purpose of the mapping.

Code	Criterion
IC1	Reports addressing at least one environmental impact in the HPC context
IC2	Reports presenting methodologies for predicting or measuring the environmental impacts of HPC systems
EC1	Reports not related to the study scope
EC2	Reports focusing solely on energy without connecting to broader environmental impacts
EC3	Reports not in English, unavailable, or inaccessible

Table 2.3: Selection criteria employed in this study.

Screening these records paired a manually reviewed baseline with an automated pass. We first screened a subset by hand and extended it through backward and forward snowballing. Three large language models then voted independently on each record of the full corpus, and we decided by hand on the records they agreed to include. The categories of the classification were not fixed in advance. We built them inductively by keywording the abstracts of the included papers and grouping the terms that recurred. The full protocol, the database exports, the prompts, and the coding files are released with the study (Rosa *et al.*, 2026). The search closed on 11 April 2026 and left a final corpus of 62 reports covering sixteen years of research.

2.3 A Field Published Elsewhere

One property of that corpus is worth stating before the classification, because it shapes how the field sees itself. More than seventy percent of the reports appeared outside the core venues of the HPC community, in general computer science, in parallel and distributed systems, and in domain-specific journals.

The corpus is also young and uneven in time (Figure 2.1). Publication was sparse and intermittent from 2010 to 2021, only seventeen reports across twelve years, with several years contributing none at all. It then expanded sharply, with four reports in 2023, twelve in 2024, and twenty-four in 2025. The year 2026 is left out, because the search reached only its first months.

Where that growth landed matters as much as its size. Between 2023 and 2025, venues outside HPC took the largest share of it, ahead of both the HPC venues and the parallel and distributed ones. Much of the recent work therefore circulates where HPC researchers do not routinely look. The distance between where this research is produced and where it would consolidate is widening rather than closing.

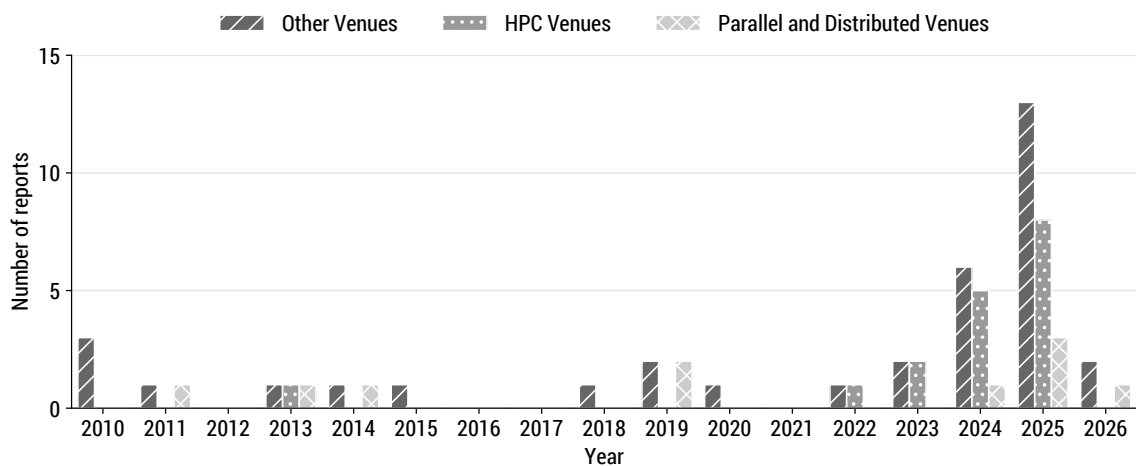


Figure 2.1: Annual report counts in the final corpus, segmented by venue category.

2.4 A Four-Facet Map of the Field

The keywording settled on four facets, and they carry the whole analysis (Figure 2.2). Each captures a different aspect of a paper. Environmental impacts identify the burden studied and divide into carbon emissions, materials and waste, water use, and biodiversity and ecosystems. Lifecycle stages locate that burden in the system's life, ordered from production, through operational use, to end of life. System loci place the work within the stack, ordered from facility and energy infrastructure, through compute hardware, to software and workloads. Management and intervention levers describe how the impact is handled, from measurement and accounting toward workload and runtime optimization, infrastructure design, and incentives and policy. Three of the four facets are ordered rather

than nominal, so a paper's place within them reads as a position on a scale, upstream to downstream, hardware to software, or measuring a problem to acting on it.

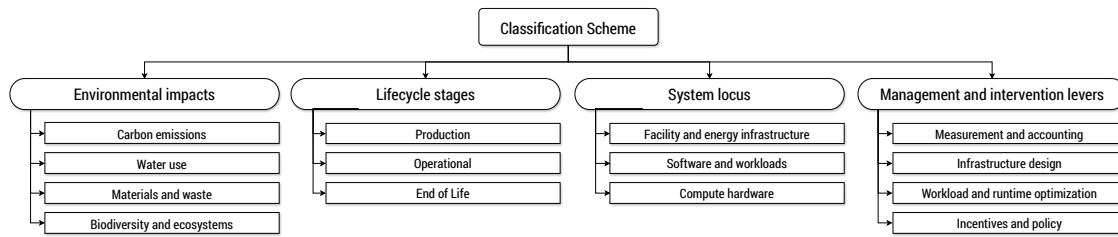


Figure 2.2: The four-facet classification scheme derived from keywording the corpus. Each facet is subdivided into categories: environmental impacts, lifecycle stages, system loci, and management and intervention levers.

Environmental impacts anchor the other three. Reading each of the remaining facets against it turns the classification into the literature map and shows which impact is studied at which stage, at which locus, and under which lever. Seen this way, the corpus has one dense region, and what the map reveals is how narrow that region is and how much of it stays empty.

2.5 The Corridor

The four impact categories are not studied in anything close to equal measure (Figure 2.3). Carbon emissions appear in 54 of the 62 reports. Materials and waste appear in 12, water use in 5, and biodiversity and ecosystems in a single report. The corpus also treats these burdens one at a time. Fifty-two reports address exactly one impact category, ten address two, and none reach three. The pairs that do occur are not arbitrary. Nine of the ten two-impact reports combine carbon with materials, which share the same manufacturing inventories, and the one remaining pair joins carbon with water through cooling.

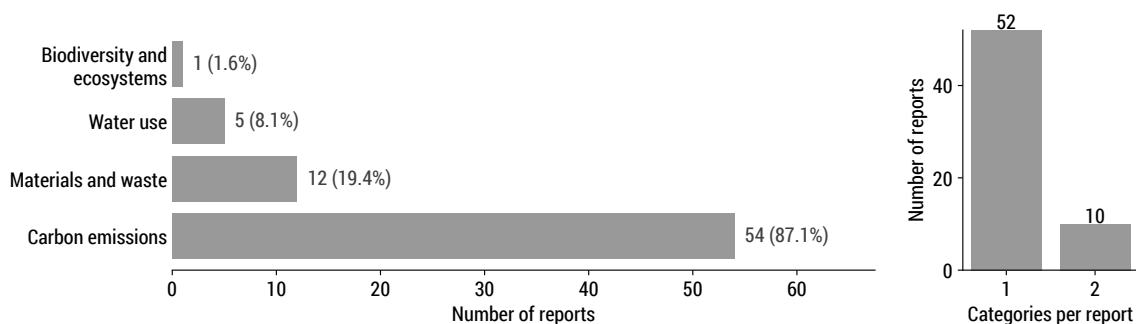


Figure 2.3: Distribution of the 62 reports across the four environmental impact categories (left) and the number of categories each report addresses (right).

The same concentration holds along the other three facets, and it lines up into a single corridor (Figure 2.4). The operational stage is the most populated point of the lifecycle for every impact category. Within carbon, the facility and energy infrastructure locus carries the most reports, 38 of them, ahead of software and workloads and of compute hardware.

Among the levers, measurement and accounting leads. The densest cells therefore share the same four coordinates: carbon, the operational stage, the facility level, and measurement. The bulk of the literature clusters where they meet.

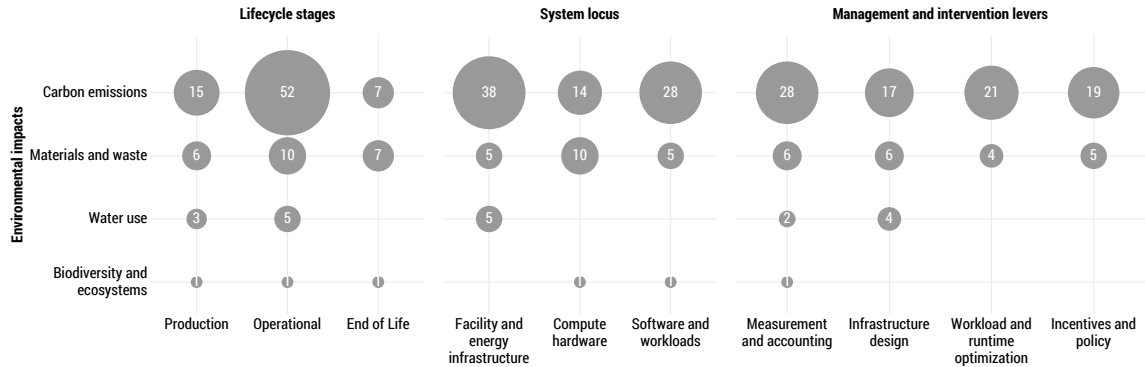


Figure 2.4: Co-occurrence of the environmental impact categories (rows) with the lifecycle stages, system loci, and management and intervention levers (columns), across the 62 reports. Bubble size encodes the number of reports in each cell, and a missing bubble marks a combination no report addresses.

Why the corridor falls where it does has less to do with which impacts matter and more with which are easy to instrument. Carbon is the readiest to quantify, because power telemetry and grid emission factors are widely available and external reporting gives an added reason to publish the numbers. Hölbling et al. (HÖLBLING *et al.*, 2025) use exactly these inputs to measure operational and embodied carbon across five Austrian universities and find a hundredfold variation in emissions per CPU hour with the regional electricity mix, and Rao et al. (RAO and CHIEN, 2025) model the carbon footprint of hundreds of Top500 systems to produce the first fleet-wide estimates. The operational stage produces continuous telemetry and job logs without extra instrumentation, the facility level offers meters and emission factors as a ready-made signal, and measurement is the lever the others build on. The corridor, in other words, traces the availability of data more than the shape of the problem.

2.6 The Periphery

Outside the corridor each minor category thins in its own way, and the empty cells are as telling as the full ones. Water use grew out of cooling engineering and has stayed there. Its five reports sit entirely at the facility level, from groundwater cooling models (SHELDON *et al.*, 2015) to thermosyphon designs that cut on-site consumption (SICKINGER *et al.*, 2018), and only recently has the category begun to treat water as a footprint to model in its own right rather than a byproduct of cooling, accounting for the indirect water spent generating electricity and weighting it by regional scarcity (JIANG *et al.*, 2025). Water reaches neither compute hardware nor software, and it never appears at end of life, even though chip fabrication consumes large volumes of ultrapure water (SANDHU *et al.*, 2025) and disposal-stage leaching can contaminate groundwater (HOUESSIONON *et al.*, 2021). Those burdens are real and documented, but not inside any HPC-specific study.

Materials and waste is the larger of the minor categories, yet it rarely stands on its own. Nine of its twelve reports also account for carbon, and the work concentrates at compute hardware because both impacts draw on the same manufacturing-stage inventories, so materials tend to ride on data already gathered for carbon accounting. Only a few papers make materials the primary object, among them a proposal for a metal usage effectiveness metric defined per unit of computation (REINBIGLER and MARTIN, 2025). Materials do reach the end of life stage more often than the other minor categories, because electronic waste creates a tangible disposal pathway. Estimates put HPC e-waste disposal costs in the hundreds of millions of dollars within the decade (ZHAO *et al.*, 2025), and several reports try to defer that cost by extending hardware life, whether by repurposing decommissioned machines (LYNAR *et al.*, 2010) or by matching replacement cycles to the balance between embodied and operational carbon (UWIZEYIMANA and JERGER, 2025).

Biodiversity and ecosystems is the extreme case, present in a single report, a fab-to-grave analysis that measures embodied and operational biodiversity impact in species-year units and finds the operational share, driven by grid electricity, far larger than the embodied one (SHI *et al.*, 2025). The methods to produce a second such study exist in the broader environmental literature, including characterization factors that translate resource use into species loss (SCHERER *et al.*, 2023), but they have not been imported into HPC. The pattern across all three minor categories is the same. The boundary of the literature traces what the community has so far built the instruments to measure, rather than the boundary of HPC's environmental footprint.

2.7 From Efficiency to Sufficiency

Concentrated on carbon measured during operation, the corpus treats efficiency as the sustainability problem, and that framing has a known limit. Gains in efficiency are repeatedly absorbed by growth in demand. Studies of digital systems document rebound effects large enough to cancel the savings (PENG *et al.*, 2023), and analyses of Top500 machines show total electricity consumption still rising even as performance per watt improves (BENHARI, TRYSTRAM, *et al.*, 2024; BENHARI, DENNEULIN, *et al.*, 2025). Efficiency also leaves whole dimensions unaccounted for. Embodied and other Scope 3 emissions are estimated at roughly 38 to 69 percent of a data center's carbon footprint (SCHNEIDER ELECTRIC, 2023), yet they sit outside the performance-per-watt metrics that dominate HPC evaluation, together with the water and material burdens the corridor never reaches.

Sufficiency concerns the scientific value of a computation relative to its environmental cost, not only the efficiency with which it runs, and requires metrics such as emissions per validated result rather than per unit of work (PORTEGIES ZWART, 2020; LANNELONGUE *et al.*, 2021). Groundwork already exists, including the Software Carbon Intensity specification with its functional units (GREEN SOFTWARE FOUNDATION, 2024) and the frugality frameworks emerging in adjacent sectors (ASSOCIATION FRANÇAISE DE NORMALISATION, 2024).

The gap exposed by the classification can be viewed from two angles. The first concerns the operational side of HPC, where the literature provides no mechanism for acting on environmental costs in scheduling, allocation, and replacement decisions. The second concerns HPC systems as a whole, which the field evaluates through efficiency metrics

where sufficiency is also needed. These angles define the two research directions pursued in the rest of this document.

Chapter 3

Preliminary Results

This chapter reports the first results of the operational branch, two studies of job timing inside the batch scheduler. The first tests a rule that suppresses backfill starts on a relative grid signal. The second prices candidate start times in absolute impact units and uses an offline analysis to bound what job shifting can achieve. Together, the studies show why refusing starts is not enough and how much remains available to a scheduler that chooses them.

3.1 Suppressing Backfill Starts on a Relative Signal

The batch scheduler is a practical place to act on operational impacts because it already decides when and where every job runs, but an environmental gain from scheduling is only worth having if the performance penalty stays acceptable. EASY backfilling is a well-established baseline for weighing that trade-off. It keeps a capacity reservation for the job at the head of the queue and raises utilization by filling the resulting gaps with later jobs that do not delay that reservation (SKOVIRA *et al.*, 1996; MU’ALEM and D. FEITELSON, 2001). The underlying question is whether a simple scheduling heuristic can lower an HPC center’s carbon or water footprint without hurting performance. Greenfilling is a first attempt at answering it, and the smallest intervention we could make against the EASY baseline. It leaves the reservation untouched and modifies exactly one decision, permitting a backfill start only when the targeted grid intensity, carbon or water, observed at the scheduling instant does not exceed its exponential moving average. Building and evaluating it doubled as a first exercise of the simulation infrastructure this branch needs.

The rule is event-driven. Greenfilling is implemented as a Batsched variant that subclasses the EASY implementation it is compared against, and the scheduler runs at every job arrival and completion. At the k -th such event, at time t_k , it reads the targeted intensity $I(t_k)$ from the 15-minute trace and updates a smoothed reference

$$S_k = \alpha I(t_k) + (1 - \alpha) S_{k-1}, \quad S_0 = I(t_0), \quad (3.1)$$

with smoothing factor $\alpha = 0.5$ throughout the campaign. The scheduler permits backfill starts at t_k if and only if $I(t_k) \leq S_k$, and it schedules the reservation for the job at the

head of the queue regardless of this condition. Because Equation (3.1) advances once per scheduling event rather than once per trace step, the effective memory of S_k tracks the density of arrivals and completions instead of wall-clock time. When many events fall inside one 15-minute step the reference converges toward the current sample, so how long backfill stays suppressed after an upward step of the signal depends on how busy the machine is, not only on α .

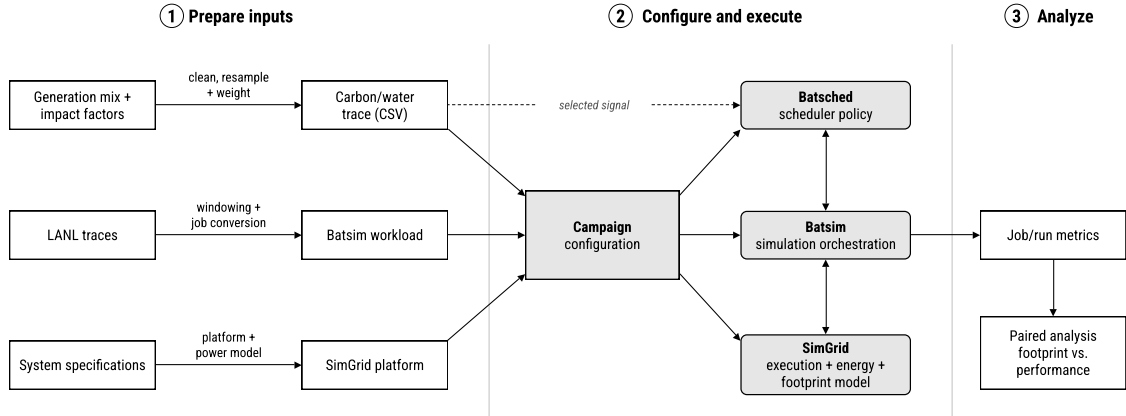


Figure 3.1: Simulation testbed for the Greenfilling campaign. Preparation converts raw sources into the three artifacts a run consumes, the campaign configuration binds them into a single simulation, and Batsched, Batsim, and SimGrid exchange decisions and events while the footprint model accumulates impacts. The dashed path marks the environmental trace reaching the scheduler directly.

Evaluating Greenfilling first required simulation support that existing toolchains do not offer, because current simulators do not track time-varying carbon and water footprints and a recent extension in this direction covers only operational carbon emissions (SARAIVA *et al.*, 2025). We therefore extended SimGrid (CASANOVA *et al.*, 2025) and Batsim (DUTOT *et al.*, 2017) with a per-host footprint model that accumulates operational carbon from the grid carbon intensity, on-site water through the water usage effectiveness (WUE), off-site water through the power usage effectiveness (PUE) and the grid water intensity, and embodied carbon and water as separate values. All four operational parameters can vary independently over time, Batsim replays timestamped updates to them during scheduling experiments, and each job’s operational and embodied footprints are exported separately.

The grid signals derive from ENTSO-E actual net generation by production type (HIRTH *et al.*, 2018), resampled to a 15-minute grid and converted into generation-weighted carbon and water intensities using lifecycle carbon factors from UNECE where available and IPCC otherwise (UNITED NATIONS, 2022; INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE, 2015) and operational water consumption factors (MACKNICK *et al.*, 2012). Germany, France, and Poland provide three contrasting signals from the same source and observation period, dominated by hard coal and lignite in Poland, by nuclear in France, and by a more mixed and changing supply in Germany. The contrast is not one-dimensional. France has the lowest mean carbon intensity of the three but the highest mean water intensity, 36.4 gCO₂eq/kWh and 3.320 L/kWh against Poland’s 569.4 gCO₂eq/kWh and 1.587 L/kWh. A consistency check against Ember’s independent production-based national intensities for 2018 to 2024 yields Pearson correlations between annual means of 0.987 for Germany,

0.994 for France, and 0.999 for Poland, with modeled levels 16.9%, 34.0%, and 19.7% lower, a gap that the choice of emission factors largely explains. Because Greenfilling reacts to changes over time rather than to absolute levels, the preserved ordering and annual trend matter more here than the level offsets.

<i>Varied factors</i>	
Grid signal country	Germany, France, Poland
Environmental window	12 per country, 3 per season, 28 days
Machine	Mustang (1,600 nodes), Trinity (9,408 nodes)
Workload regime	slack, stress
Scheduler	EASY, carbon Greenfilling, water Greenfilling
<i>Fixed settings</i>	
EMA smoothing factor	$\alpha = 0.5$
Grid signal resolution	15 minutes
Node power	10 W idle, 320/270 W busy (Mustang/Trinity)
Facility model	PUE = 1, WUE = 0, no embodied impacts
Total	432 runs, 288 paired comparisons

Table 3.1: *Parameters of the Greenfilling evaluation campaign.*

The evaluation campaign pairs these signals with two LANL machines of different scales, Mustang at 1,600 nodes and Trinity at 9,408 nodes, each represented by one slack and one stress workload extract of four weeks selected on utilization and queue conditions. The platform model is deliberately simple: jobs are rigid and replayed at their recorded size, nodes draw 10 W idle and a constant 320 W or 270 W while busy on Mustang and Trinity, and the facility defaults to a PUE of 1, a WUE of 0, and no embodied values, so the campaign reports operational grid carbon and off-site water alone. Constant active power also bounds what scheduling can reach, because reordering the same jobs cannot change active energy and the only remaining levers are idle periods and the operational intensities a job encounters while it runs. Each country contributes 12 environmental windows of 28 days, three drawn per season under a fixed seed, for 36 windows in total. [Figure 3.1](#) shows the final experimental testbed architecture, while [Table 3.1](#) summarizes the parameters of the campaign.

Crossing windows, machines, workload regimes, and three schedulers, EASY plus a carbon-targeting and a water-targeting Greenfilling, gives 432 simulations, and pairing each Greenfilling run with the EASY run of the same window, machine, and regime gives 288 paired comparisons. Greenfilling is not competitive in any evaluated scenario, that is, it does not reduce the targeted operational footprint relative to EASY without increasing mean bounded slowdown. [Figure 3.2](#) shows the paired changes relative to EASY, and [Table 3.2](#) breaks the medians down by machine, regime, and objective. No scenario median footprint change reaches 0.1% in magnitude, for carbon or for water, while mean bounded slowdown rises in every one of the 288 pairs, between +24.5% and +757.2%, with scenario medians between +161.5% and +323.6%. The 95th-percentile waiting time of replayed jobs also rises in all 288 pairs. Even the most favorable case, Mustang under the slack regime,

Machine	Regime	Objective	Footprint	Energy	Slowdown
Mustang	slack	carbon	−0.068%	0.000%	+323.6%
		water	−0.031%	0.000%	+308.5%
	stress	carbon	+0.008%	+0.043%	+175.0%
		water	+0.014%	+0.044%	+161.5%
Trinity	slack	carbon	−0.005%	0.000%	+210.1%
		water	−0.007%	0.000%	+211.5%
	stress	carbon	+0.001%	0.000%	+226.6%
		water	−0.000%	0.000%	+213.3%

Table 3.2: Median paired changes relative to EASY by machine, workload regime, and Greenfilling objective. The footprint column reports the signal each objective targets. On Mustang under slack the consumed platform energy is identical to EASY in every window, and on Trinity it is unchanged to numerical precision at the median.

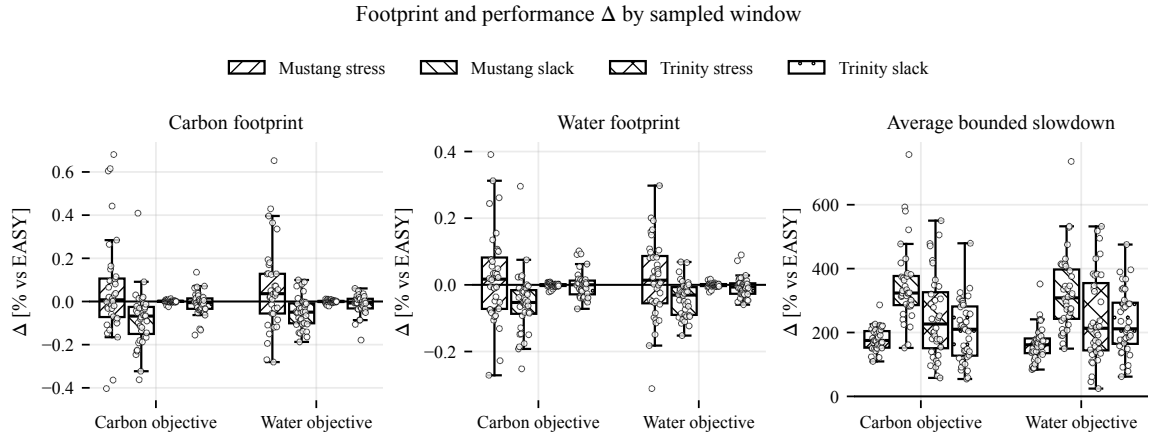


Figure 3.2: Footprint and performance change relative to EASY across the 288 paired comparisons. Carbon and water footprint changes stay small, with medians below 0.1% in magnitude, while mean bounded slowdown worsens in every pair.

where carbon falls in 30 of 36 windows under carbon Greenfilling and water falls in 29 of 36 under water Greenfilling, buys those reductions at a median slowdown increase of +323.6%. Under stress the Mustang medians turn slightly positive and Trinity stays near zero in both regimes. Zero of the 288 pairs reduce the targeted footprint without increasing mean bounded slowdown.

The energy column of Table 3.2 separates the two ways a footprint can change under this platform model, consuming a different amount of energy or meeting a different intensity while consuming it. On Mustang under slack and on Trinity in both regimes the consumed energy does not move, so the footprint change there is purely an exposure change, and that exposure change stays below 0.1%. Mustang under stress is the one exception. Median platform energy rises by 0.043% under carbon Greenfilling and 0.044% under water Greenfilling, while the effective targeted intensity, the targeted operational footprint divided by consumed energy, falls by 0.037% and 0.032%. The rule does buy a small exposure improvement there, but the extra energy from stretching the schedule

cancels it, which is how the stress medians on Mustang end up slightly positive. The campaign accounts energy at the whole-platform level, so this decomposition shows the mechanism but cannot attribute the extra energy to individual jobs.

The design itself suggests why deferral did not translate into lower footprints. Greenfilling decides whether a backfill may start at the current instant, but it never selects a later start based on predicted intensity. Under constant active power, postponing a job changes its footprint only if its execution actually moves to lower-intensity intervals, and the campaign does not determine whether that condition failed because of the decision rule, the temporal structure of the signals, the workloads, or their interaction. The campaign does probe the signal hypothesis once. Each window’s swing, the mean daily peak-to-trough range of the targeted signal divided by the window mean, correlates only weakly with the achieved saving, at a pooled Pearson $r = 0.079$ for carbon and $r = 0.247$ for water over 144 pairs per objective, with per-scenario coefficients between -0.09 and 0.59 and no consistent sign. Every window is reused by all four machine and regime scenarios, so the pooled points are not independent, but windows with more intra-day variation did not systematically yield larger savings. The result is therefore narrow. Under the evaluated signals, workloads, and platform assumptions, suppressing EASY’s backfill starts on a moving-average comparison does not reduce the targeted footprint enough to offset its increase in mean bounded slowdown.

Two lessons from the campaign now shape the mechanism that replaces it. The first is the asymmetry between refusing a start and choosing one. A suppressed backfill start trades a certain queueing delay for an environmental return that depends on the intensities the deferred job eventually meets, and the campaign shows how one-sided that trade is. The second is that a relative threshold and a relative metric compound each other. The rule fires on the same relative deviation in France as in Poland even though their mean carbon intensities differ by more than an order of magnitude, a 0.1% saving reads identically in both, and the quantity that would distinguish them, the absolute footprint avoided per unit of scheduling penalty, was not measured. Together, these lessons set a lower bound. An intervention that only refuses starts on a relative signal is too little. A viable heuristic has to price both sides of each decision, the footprint of starting a job now against the footprint of waiting, and it has to weigh absolute rather than relative stakes.

The campaign also produced reusable infrastructure. We release the intensity traces, workload extracts, campaign configuration, and analysis as a reproducible artifact,¹ and we submitted this analysis to the EESP 2026 workshop.

3.2 Pricing Each Start in Absolute Terms

The Greenfilling lessons point to job shifting as the next mechanism to explore. Rather than suppressing opportunistic starts on a relative comparison, the scheduler could choose each job’s start time by pricing every candidate start within a bounded delay, in the absolute units of the targeted impact.

The model prices a single job: let j be a job with execution time p_j , q_j allocated nodes,

¹ <https://github.com/Lucas-Doctorate-Project/experimental-artifacts/tree/paper>

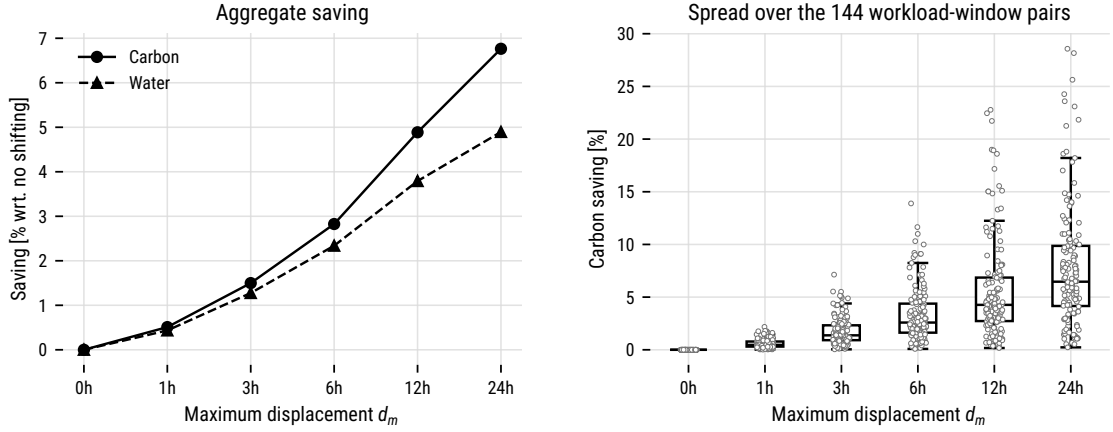


Figure 3.3: Ceiling of the offline job-shifting analysis as a function of the maximum displacement d_m . The left panel shows the aggregate saving over all workloads and windows for each signal. The right panel shows the spread of the carbon saving across the 144 workload and window pairs.

Grid	Signal	Mean	Median	Std	Min	Max
Germany	carbon	0.50	0.45	0.22	0.29	0.93
	water	0.47	0.39	0.21	0.27	0.84
France	carbon	0.32	0.30	0.08	0.18	0.51
	water	0.29	0.27	0.07	0.19	0.44
Poland	carbon	0.24	0.20	0.15	0.09	0.61
	water	0.23	0.19	0.15	0.10	0.60

Table 3.3: Swing of the carbon and water signals by grid, over the 12 windows per country used in the Greenfilling campaign.

and submission time r_j , and let I be the time-varying intensity of the targeted signal. Starting j at a time $t \geq r_j$ costs

$$C_j(t) = P_{\text{idle}} q_j \int_{r_j}^t I(\tau) d\tau + P_{\text{comp}} q_j \int_t^{t+p_j} I(\tau) d\tau, \quad (3.2)$$

where P_{idle} and P_{comp} are the platform's idle and busy node powers. The first term charges for the wait. While j holds back, its q_j nodes sit idle and accumulate impact at the intensities of the interval $[r_j, t)$. The second term charges for the run, integrating the intensity over the window the execution would actually occupy rather than sampling it at the start. The heuristic selects the start t^* that minimizes C_j over $r_j \leq t \leq r_j + d_m$, where the maximum displacement d_m bounds how long any job can be held. At $t = r_j$ the waiting term vanishes and Equation (3.2) reduces to the cost of starting immediately, so not shifting is itself a candidate and the heuristic only moves a job when the trace ahead pays for the wait. Carbon and water each define their own cost function through their own intensity series, and thus their own t^* . The cost of deferral appears as idle exposure, the return appears as the difference in integrated intensity over the execution window, and both sides are absolute quantities rather than deviations from a moving average.

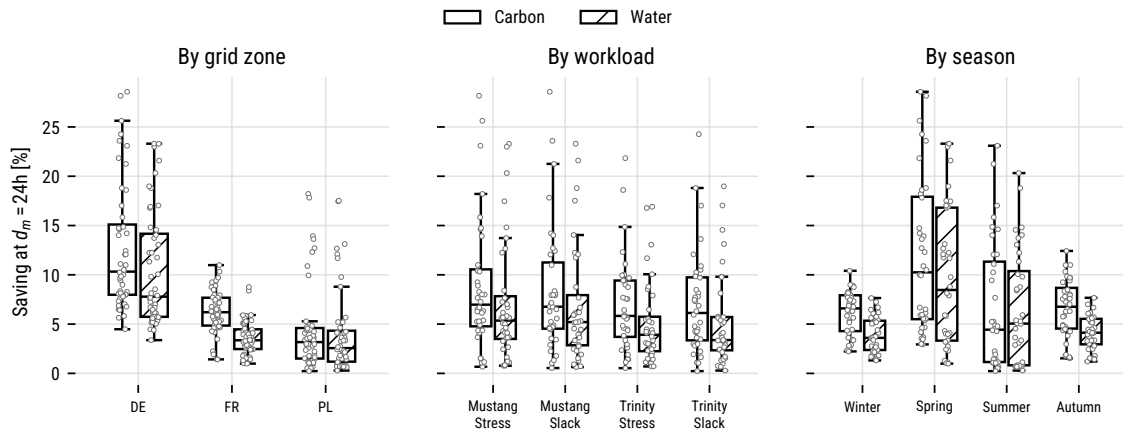


Figure 3.4: Ceiling at a d_m of 24 hours across the 144 workload and window pairs, split by grid zone, workload scenario, and season.

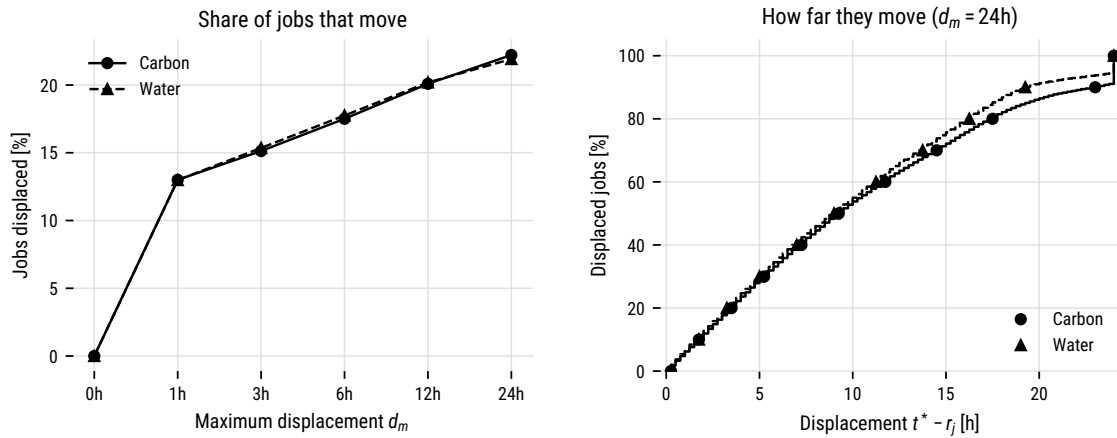


Figure 3.5: Share of jobs the offline analysis displaces as a function of d_m (left) and the distribution of the displacement $t^* - r_j$ among displaced jobs at a d_m of 24 hours (right).

Before the heuristic reaches the scheduler, the same cost function supports an offline analysis of the workloads and traces. The grid signals are step functions on a 15-minute grid, so the integrals in Equation (3.2) collapse into sums. We use this to compute a ceiling: for each job, excluding the warm-up jobs that reconstruct the machine state at the start of each extract, we record the saving it would collect if it could shift freely within d_m , and we accumulate these per-job gains over the same 144 workload and window combinations as the campaign, with the same platform power values, for d_m between zero and 24 hours. The ceiling is optimistic by construction, because summing independent per-job gains ignores node capacity and queue contention and lets every job claim the same low-intensity periods at once. Its value is diagnostic. If even this bound were small, no shifting heuristic on these signals would be worth implementing.

The bound is not small, as Figure 3.3 shows. The aggregate ceiling grows with the allowed displacement and reaches 6.76% for carbon and 4.89% for water at a d_m of 24 hours, almost two orders of magnitude above the sub-0.1% medians of the Greenfilling campaign. Water saves less than carbon, likely because of its lower day-to-day variability.

Averaged over the 12 windows per country used in the Greenfilling campaign, mean swing is 0.50 for carbon against 0.47 for water in Germany, 0.32 against 0.29 in France, and 0.24 against 0.23 in Poland (Table 3.3).

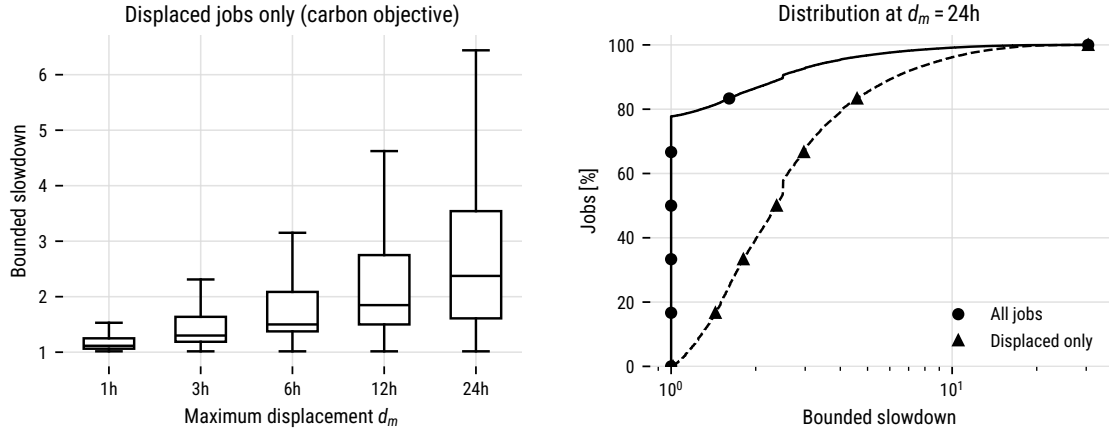


Figure 3.6: Bounded slowdown implied by the displacements under the carbon objective. The left panel shows displaced jobs only, by maximum displacement. The right panel shows the cumulative distribution at a d_m of 24 hours for all jobs and for displaced jobs only.

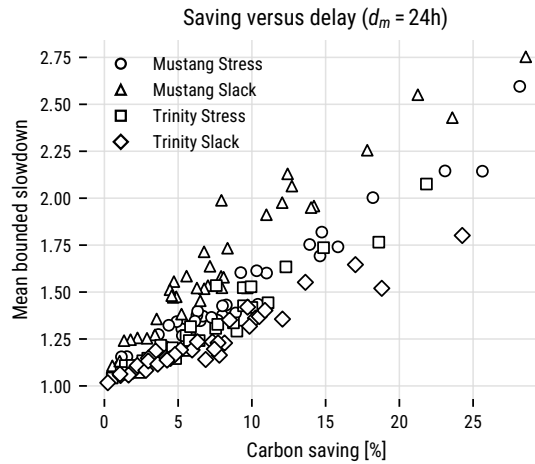


Figure 3.7: Carbon saving against mean bounded slowdown at a d_m of 24 hours, one point per workload and window pair.

Within that bound, the gains concentrate by grid and by season (Figure 3.4). The German signal offers the most, with an aggregate ceiling of 11.69% at 24 hours, against far less in France and Poland, and the seasons follow the same mechanism. Spring reaches 12.32% against winter's 4.89%, and the gap tracks the same swing metric. Over the three windows drawn from each of the three grids, mean carbon swing is 0.50 in spring against 0.28 in winter, nearly double, so a job displaced in spring meets a wider daily range between low and high intensity than one displaced in winter, and the heuristic exploits that wider range. The four workload scenarios behave similarly, which points at the signal rather than the workload as the driver, though with only three windows per grid and season the seasonal split is still a tendency rather than a settled result.

Only a fraction of jobs actually move (Figure 3.5). At a d_m of 24 hours, 22.2% of jobs are displaced at all, with a median displacement of 9.25 hours among those that move, while the remaining 77.8% already start at the best point they can reach. The aggregate ceiling is therefore carried by roughly a fifth of the jobs, which sharpens the capacity caveat, because the bound assumes all of them fit into the same periods at the same time.

Shifting a job deterministically adds $t^* - r_j$ to its waiting time, so its bounded slowdown, $\max(1, (t^* - r_j + p_j) / \max(p_j, \tau))$ with a bounding threshold of $\tau = 10$ s, rises with every displacement. Under the carbon objective at a d_m of 24 hours the mean bounded slowdown is 1.49 across all jobs and 3.22 among displaced jobs, with a 99th percentile of 15.4 (Figure 3.6). Saving and delay also move together across scenarios. Per workload and window pair, the carbon saving correlates with the mean bounded slowdown at $r = 0.874$ (Figure 3.7). The windows that save the most are the windows where jobs wait the longest, so the trade-off is still there, but unlike under Greenfilling each unit of delay now buys a measured environmental return.

Chapter 4

Research Plan

This chapter presents the research plan through the thesis defense. [Section 4.1](#) presents what to expect from the job shifting experiments and how we plan to develop mechanisms involving the scheduler, users, platform controls, and hardware replacement. [Section 4.2](#) presents the sufficiency-aware evaluation of public machine data.

4.1 Accounting for Environmental Cost in HPC Operations

Greenfilling showed that a threshold rule alone is not enough. Refusing a start when the current intensity sits above its moving average trades a certain delay for a return that depends on intensities the deferred job may never meet. [Chapter 3](#) reports the outcome: no scenario median moved the targeted footprint by 0.1%, and mean bounded slowdown rose in all 288 pairs. A policy therefore has to price both sides of the decision, the impact of acting now against the impact of waiting.

The operational cost of an HPC center is shaped by four entities, shown in [Figure 4.1](#). Users typically submit work and are charged in core-hours. The scheduler chooses when and where each job runs, based on queue state and node availability. The platform runs its nodes at fixed power and frequency settings. Hardware is often replaced on a cycle determined by performance and procurement considerations. Environmental cost rarely enters any of these decisions.

The rest of this section brings environmental cost into each of these decisions, one step at a time. The first step stays with the scheduler and completes the job shifting study started in [Chapter 3](#). The second step brings in the platform, turning its power settings into another variable the scheduler can price alongside start time. The third step moves past day-to-day operation into hardware aging and replacement. The fourth step reaches the users, who decide what work arrives and how much delay it tolerates.

Job shifting is already known in the literature. Wiesner et al. ([WIESNER *et al.*, 2021](#)) report savings from temporal shifting in the cloud, and Sukprasert et al. ([SUKPRASERT *et al.*, 2024](#)) report how quickly those savings shrink once deadlines and workload structure

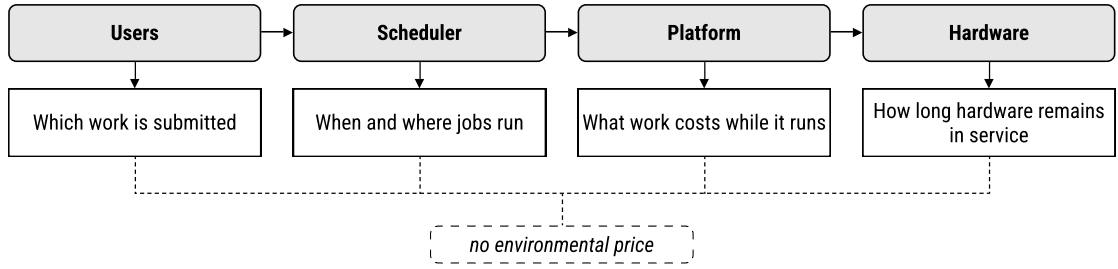


Figure 4.1: *Entities that dictate the operational cost of HPC. Users decide what work is submitted, the scheduler decides when and where it runs, the platform sets what running it costs, and hardware replacement decides how long the machine stays in service. These decisions rarely price the environmental cost.*

are respected. Equation (3.2) prices a single job in isolation, and the offline analysis of Chapter 3 used it to bound what shifting can reach, 6.76% of operational carbon and 4.89% of water at a maximum displacement of 24 hours. That bound is unrestricted in two ways. It lets every one of the 22.2% of displaced jobs claim the same low-intensity periods at once, and it assumes the whole intensity trace is known at the moment of the decision. Implementing the cost function as a Batsched scheduler removes the first assumption, because node capacity then decides which jobs actually reach the periods they price. We will evaluate it against EASY and both Greenfilling variants on the campaign design of Table 3.1, sweeping d_m . We will then report what fraction of the offline bound survives contention and what that fraction costs in bounded slowdown.

The next question is how much of the realized saving survives when the cost function uses a forecast instead of the recorded signal. The ENTSO-E platform that supplies the campaign signals also publishes day-ahead generation forecasts (HIRTH *et al.*, 2018), so the data is readily available. This removes the second assumption and answers the questions left open in Chapter 3.

The second step starts from a limitation of the platform model. It holds active power constant, so reordering the same jobs cannot change active energy. Only idle periods and the intensities a job meets while it runs remain available to the scheduler. Other controls exist but were not explored, namely node power states and per-node DVFS and RAPL power caps (DAVID *et al.*, 2010). EcoFreq (KOZLOV and STAMATAKIS, 2024) scales power against carbon and price signals, and gives us a reference point.

Equation (3.2) currently prices only a job’s start time. Adding the power state as a second decision variable lets the scheduler price both together, trading execution time against power as well as against delay. This raises a question the earlier steps cannot ask: does slowing a job down beat making it wait? Running at a lower frequency stretches a job across a longer window of the signal and lowers its draw throughout, while waiting preserves the draw and only moves it. Neither option is obviously better. Which one wins should depend on the shape of the signal and on how a job’s runtime responds to frequency.

Cooling offers a second set of controls beyond the compute nodes. The footprint model already supports time-varying PUE and WUE, but the Greenfilling campaign kept these values fixed and left the capability unused. Making cooling a decision variable would

require a thermal model that the testbed currently lacks. We will prioritize compute controls first, and investigate cooling controls only if the results show that the trade-off between execution time and power consumption justifies the additional model complexity.

The third step leaves the operational stage for replacement timing. This decision determines how much manufacturing impact a center commits to and how long it has to amortize that impact. Li et al. (LI *et al.*, 2023) and Uwizeyimana and Enright Jerger (UWIZEYIMANA and JERGER, 2025) frame it as a crossover. A newer machine lowers operational impact per unit of work while adding embodied impact at once, so replacing early spends carbon to save carbon, and the balance depends on how the machine is used.

The crossover weighs the embodied impact of a new machine against the operational impact of keeping the current one running. Prior work already estimates the latter for a fixed workload. We want to adapt this idea to footprint-aware scheduling, delaying the crossover and extending the machine's service life when the trade-off justifies it. Scheduling, usually treated as an operational concern, then becomes part of a lifecycle decision.

The testbed already models embodied carbon and water, so a replacement policy can reason directly about these impacts, amortized over the useful work the machine delivers. Because replacement decisions play out over several years, we will also need a long-horizon simulation setup that captures hardware aging.

The fourth step makes the user an active participant. The user declares how much delay a job tolerates, whether it accepts a reduced power state, and whether the job is submitted at all. Kamatar et al. (KAMATAR *et al.*, 2025) surveyed 300 HPC users and found that fewer than 30% were aware of the energy their jobs consume. Work on parallel system traces established that submission behavior responds to how the system performs (SHMUELI and D. G. FEITELSON, 2007). Other works attribute energy and emissions to individual jobs and users, either through resource manager monitoring stacks (PAIPURI, 2025) or through user-centric footprints computed from job records (WASSERMANN *et al.*, 2024; NIEWENHUIS *et al.*, 2024). The Fugaku Points program goes further and rewards users for applying power-control settings (SOLÓRZANO *et al.*, 2024).

At this stage the platform can also respond to the user with a price, a reward, or both, and we want to understand how user decisions change when such a policy is introduced. Price and reward can be independent mechanisms. A reward can carry no budget of its own, or it can be funded by what the price collects, so that users who defer are paid by users who do not.

We will compare mechanisms on the footprint they remove and on the cost they impose on users in delay and refused work. Because no trace records how users respond to a price, we will model synthetic users and their behavior, which lets us evaluate different scenarios and identify which factors dominate in each case.

Together, the four steps cover interactions that exist in real systems. The scheduler places work, the platform sets what that work costs, and users decide what work arrives and how much delay it tolerates. The goal of this branch is to understand how these entities interact.

4.2 Sufficiency-Aware Evaluation of HPC Systems

The mapping study in [Chapter 2](#) showed that evaluation in the field is focused on operational efficiency. The Green500 ranks supercomputers by performance per watt, measured on a single benchmark ([FENG and CAMERON, 2007](#)). That metric gave the field a shared target beyond raw speed, and the list has run for fifteen years on it ([ADHINARAYANAN and FENG, 2025](#)). However, a ratio of one impact to one measure of useful work leaves most of a system's footprint unaccounted for. Karanikolaou and Bekakos ([KARANIKOLAOU and BEKAKOS, 2022](#)) find systems of entirely different size and capability sitting on consecutive rows of the Green500. Rao and Chien ([RAO and CHIEN, 2025](#)) put numbers on the same effect in the Top500. Leonardo emits 4.3 times the operational carbon of LUMI, the system ranked directly above it, and Frontier carries 2.6 times the embodied carbon of El Capitan. Absolute power draw, hardware composition, the carbon intensity of the local grid, and the water spent generating that electricity all separate those machines, and a performance-per-watt ratio reports none of it.

Our aim is to use the ranking as a reference point and show what a sufficiency-aware view reveals once that ranking is compared against the rest of a system's footprint. Efficiency measures how much useful work a system returns per unit of impact, a ratio that improves as the numerator grows. Sufficiency asks instead whether the total burden a system imposes, and the growth in demand behind it, can be reduced at all ([SANTARIUS *et al.*, 2022](#)). A relative metric can improve while the absolute quantity it divides keeps growing. That is how a fleet becomes greener per operation and heavier in total at the same time ([YORK *et al.*, 2022](#)).

The Top500 already sits in that position: performance per unit of carbon keeps improving while the fleet's operational carbon grows by 10.3% a year, on track to nearly double between 2024 and 2030 ([RAO and CHIEN, 2025](#)). The same effect appears beyond the machine room. Yang and Wang ([YANG and WANG, 2023](#)) compare a Chinese national supercomputing center against a synthetic control city and find that it worsened regional carbon emissions, because growth in energy demand outweighed what the center returned through a cleaner energy structure or greener technology. We propose a diagnostic view that exposes what efficiency-only evaluation hides, as a step towards deciding how much computation is justified.

We plan to approach this gradually, in three steps that move further from efficiency toward sufficiency. For each step we evaluate what it reveals, what data it demands, and what it trades away. [Figure 4.2](#) places the three steps against the boundary of what public data supports.

The first step keeps the shape of a ranking but widens the impacts it accounts for. Carbon models already cover both the operational and the manufacturing stages ([LI *et al.*, 2023](#)). Water can be estimated from the electricity a system draws and from the scarcity of the region supplying it ([JIANG *et al.*, 2025](#)). Instead of performance per watt alone, we will rank systems according to the impact dimensions identified by the mapping study.

The second step introduces sufficiency-oriented indicators, among them a system's total lifecycle burden, how heavily it is actually used, and how often it is replaced. These indicators are where the efficiency account and the sufficiency account diverge.

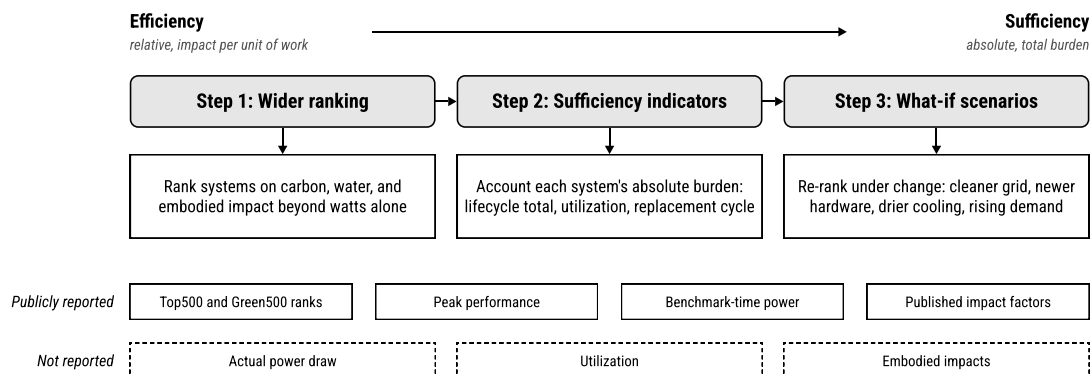


Figure 4.2: The three evaluation steps move from a wider efficiency ranking toward an account of absolute burden. The public-data boundary cuts across the whole plan rather than any single step: ranked entries, peak performance, benchmark-time power, and published impact factors are on the record, while actual power draw, utilization, and embodied impacts are not.

The third step turns the missing data into the object of study. Rather than rank systems only as they stand, we will test what the ranking becomes under a change: the same machine on a cleaner grid, with a newer generation of hardware, under a less water-intensive cooling design, or against a rising demand trajectory. Each scenario supplies an assumed figure where the real one is unavailable, then shows how the re-ranking responds.

The main challenge with these steps is data. Rao and Chien (RAO and CHIEN, 2025) found a utilization figure for three of the 500 ranked systems, and essentially none of them report actual power draw. They also show that, once public sources beyond the list are added, operational carbon can be estimated for 98% of the ranked systems and embodied carbon for 80.8%. We expect to resort to estimation and modelling. Beyond the numbers themselves, we intend to provide guidelines on how to make this information more accessible.

Taken together, the three steps move from a wider efficiency ranking toward an account of absolute burden. They rest on the same foundation as the rest of this thesis: the map of Chapter 2 and the premise that the impacts of HPC extend past energy efficiency. The Green500 gave HPC a way to reward efficient machines. We aim to add a way to account for what efficiency alone leaves out, and to make how much computing is enough part of how systems are judged.

Chapter 5

Schedule and Expected Contributions

This chapter turns the plan of [Chapter 4](#) into a timeline and a list of deliverables. [Section 5.1](#) allocates the steps of both branches over the two years between this qualification and the thesis defense. [Section 5.2](#) lists what the thesis has produced so far and what it commits to produce.

5.1 Schedule

[Figure 5.1](#) lays the steps of [Chapter 4](#) over eight quarters, from September 2026 to August 2028. The branches run in parallel throughout. The job shifting arc comes first, since [Chapter 3](#) left it one implementation away from an answer. The first step of the sufficiency branch follows one quarter later, once the shifting work is underway, and it stays near-term because it needs only public Top500 entries and published impact factors. The later steps of each branch follow the order [Chapter 4](#) gives them. Replacement timing overlaps the end of the power state work because both build on the same simulation infrastructure, and the user mechanisms close the operations branch in the second year, starting as the what-if scenarios wind down.

The two milestones in the middle of the timeline are paper submissions, one per branch. The first collects the operations results available by the end of the first year, the online shifting evaluation and the power state pricing. The second collects the wider ranking and the sufficiency indicators. We plan to submit both to high impact venues in the field, such as Supercomputing, ISC, and SBAC-PAD. The final six months are reserved for writing the thesis, which by then assembles chapters that already exist in some form as papers, analyses, and released artifacts.

5.2 Expected Contributions

[Table 5.1](#) lists the contributions of this thesis, separating what is already delivered from what the schedule above commits to. The mapping study is published in Computer Science Review ([Rosa et al., 2026](#)). The footprint model and the Greenfilling analysis exist and support everything the operations branch plans next, and the analysis is under review

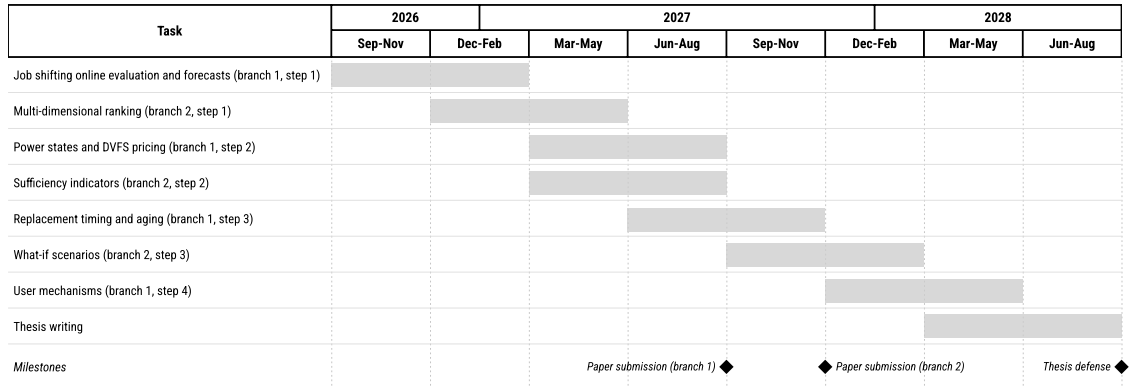


Figure 5.1: Schedule from the qualification to the thesis defense, in quarters. Each task carries the branch and step it implements from [Chapter 4](#). Diamonds mark the two planned paper submissions and the defense.

Contribution	Branch	Status
Systematic mapping study (Rosa et al., 2026), Elsevier CSR	both	published
Footprint model for SimGrid and Batsim	1	done
Job shifting analysis, EESP 2026	1	submitted
Reproducible experimental artifact	both	ongoing
Co-supervision of two undergraduate students	1	ongoing
Operations branch paper	1	planned
Sufficiency evaluation paper	2	planned
Upstreaming the footprint model into SimGrid and Batsim	1	planned

Table 5.1: Contributions of this thesis. Branch 1 is the operations branch of [Section 4.1](#) and branch 2 the sufficiency evaluation of [Section 4.2](#).

at the 3rd International Workshop on Energy Efficiency with Sustainable Performance (IEEE/ACM SC EESP Workshop 2026). The experimental artifact is ongoing work that accumulates the traces, configurations, and analyses of each campaign as the research advances. Two undergraduate students work under our co-supervision on the preliminary results line, extending the shifting analysis. Once the footprint model and the simulation infrastructure around it are stable, we plan to upstream them into SimGrid and Batsim rather than maintain a fork, so the infrastructure outlives the thesis.

These deliverables also carry contributions beyond their immediate results. For the field, the mapping study and the sufficiency evaluation widen how systems are judged beyond performance per watt, and the data boundary drawn in [Section 4.2](#) tells operators and vendors exactly which figures are missing and what publishing them would support. For practitioners, the mechanisms of the operations branch are candidate policies for schedulers and resource managers already in production, and their evaluation reports what every operator asks first, how much footprint a policy removes and how much slowdown it costs. For researchers, the footprint model, the traces, and the campaign

tooling lower the cost of entering this line of work, and upstreaming them into SimGrid and Batsim keeps that infrastructure maintained after the thesis ends.

References

- [ADHINARAYANAN and FENG 2025] Vignesh ADHINARAYANAN and Wu-chun FENG. “Looking Back to Look Forward: 15 Years of the Green500”. *Computer* 58.1 (Jan. 2025), pp. 76–86. ISSN: 0018-9162, 1558-0814. DOI: [10.1109/MC.2023.3333316](https://doi.org/10.1109/MC.2023.3333316) (cit. on p. 26).
- [ALLEN 2022] Michael ALLEN. “The huge carbon footprint of large-scale computing”. *Physics World* 35.3 (Aug. 2022), pp. 46–50. ISSN: 0953-8585, 2058-7058. DOI: [10.1088/2058-7058/35/03/32](https://doi.org/10.1088/2058-7058/35/03/32) (cit. on p. 1).
- [ASSOCIATION FRANÇAISE DE NORMALISATION 2024] ASSOCIATION FRANÇAISE DE NORMALISATION. *General Framework for Frugal AI - Measuring and Reducing the Environmental Impact of AI*. June 2024 (cit. on p. 11).
- [BENHARI, DENNEULIN, *et al.* 2025] Abdessalam BENHARI, Yves DENNEULIN, Frédéric DESPREZ, Fanny DUFOSSÉ, and Denis TRYSTRAM. *Analysis of the Carbon Footprint of HPC*. Sept. 2025. DOI: [10.48550/arXiv.2509.22679](https://doi.org/10.48550/arXiv.2509.22679). eprint: [2509.22679](https://arxiv.org/abs/2509.22679) (cit. on p. 11).
- [BENHARI, TRYSTRAM, *et al.* 2024] Abdessalam BENHARI, Denis TRYSTRAM, Fanny DUFOSSÉ, Yves DENNEULIN, and Frédéric DESPREZ. *Green HPC: An Analysis of the Domain Based on Top500*. Mar. 2024. DOI: [10.48550/arXiv.2403.17466](https://doi.org/10.48550/arXiv.2403.17466). eprint: [2403.17466](https://arxiv.org/abs/2403.17466) (cit. on p. 11).
- [CAI *et al.* 2011] Chang CAI, Lizhe WANG, Samee U. KHAN, and Jie TAO. “Energy-Aware High Performance Computing: A Taxonomy Study”. In: *2011 IEEE 17th International Conference on Parallel and Distributed Systems*. Tainan, Taiwan: IEEE, Dec. 2011, pp. 953–958. ISBN: 978-0-7695-4576-9 978-1-4577-1875-5. DOI: [10.1109/ICPADS.2011.59](https://doi.org/10.1109/ICPADS.2011.59) (cit. on pp. 5, 6).
- [CASANOVA *et al.* 2025] Henri CASANOVA, Arnaud GIERSCHE, Arnaud LEGRAND, Martin QUINSON, and Frédéric SUTER. “Lowering entry barriers to developing custom simulators of distributed applications and platforms with SimGrid”. *Parallel Computing* 123 (Mar. 2025), p. 103125. ISSN: 01678191. DOI: [10.1016/j.parco.2025.103125](https://doi.org/10.1016/j.parco.2025.103125) (cit. on p. 14).

- [CHAUDHRY *et al.* 2015] Muhammad Tayyab CHAUDHRY, Teck Chaw LING, Atif MANZOOR, Syed Asad HUSSAIN, and Jongwon KIM. “Thermal-Aware Scheduling in Green Data Centers”. *ACM Computing Surveys* 47.3 (Apr. 2015), pp. 1–48. ISSN: 0360-0300, 1557-7341. DOI: [10.1145/2678278](https://doi.org/10.1145/2678278) (cit. on pp. 5, 6).
- [CZARNUL *et al.* 2019] Pawel CZARNUL, Jerzy PROFICZ, and Adam KRZYWANIAK. “Energy-Aware High-Performance Computing: Survey of State-of-the-Art Tools, Techniques, and Environments”. *Scientific Programming* 2019 (Apr. 2019), pp. 1–19. ISSN: 1058-9244, 1875-919X. DOI: [10.1155/2019/8348791](https://doi.org/10.1155/2019/8348791) (cit. on pp. 1, 5, 6).
- [D’ORGEVAL *et al.* 2024] Alexandre D’ORGEVAL, Valentina SESSA, Edi ASSOUMOU, and Quentin AVENAS. *Towards Improved Datacenter Assessment: Review and Framework Proposition*. Jan. 2024. DOI: [10.46855/energy-proceedings-11079](https://doi.org/10.46855/energy-proceedings-11079) (cit. on p. 6).
- [DAVID *et al.* 2010] Howard DAVID, Eugene GORBATOV, Ulf R. HANEBUTTE, Rahul KHANNA, and Christian LE. “RAPL: memory power estimation and capping”. In: *Proceedings of the 16th ACM/IEEE International Symposium on Low Power Electronics and Design*. Austin Texas USA: ACM, Aug. 2010, pp. 189–194. ISBN: 978-1-4503-0146-6. DOI: [10.1145/1840845.1840883](https://doi.org/10.1145/1840845.1840883) (cit. on p. 24).
- [DUTOT *et al.* 2017] Pierre-François DUTOT, Michael MERCIER, Millian POQUET, and Olivier RICHARD. “Batsim: a realistic language-independent resources and jobs management systems simulator”. In: *Job Scheduling Strategies for Parallel Processing*. Ed. by Narayan DESAI and Walfredo CIRNE. Cham: Springer International Publishing, 2017, pp. 178–197. ISBN: 978-3-319-61756-5 (cit. on p. 14).
- [FENG and CAMERON 2007] Wu-chun FENG and Kirk CAMERON. “The Green500 List: Encouraging Sustainable Supercomputing”. *Computer* 40.12 (Dec. 2007), pp. 50–55. ISSN: 0018-9162. DOI: [10.1109/MC.2007.445](https://doi.org/10.1109/MC.2007.445) (cit. on pp. 1, 26).
- [GREEN SOFTWARE FOUNDATION 2024] GREEN SOFTWARE FOUNDATION. *Software Carbon Intensity (SCI) Specification*. <https://sci.greensoftware.foundation/>. 2024 (cit. on p. 11).
- [GUPTA *et al.* 2021] Udit GUPTA *et al.* “Chasing Carbon: The Elusive Environmental Footprint of Computing”. In: *2021 IEEE International Symposium on High-Performance Computer Architecture (HPCA)*. Seoul, Korea (South): IEEE, Feb. 2021, pp. 854–867. ISBN: 978-1-6654-2235-2. DOI: [10.1109/HPCA51647.2021.00076](https://doi.org/10.1109/HPCA51647.2021.00076) (cit. on p. 1).
- [HIRTH *et al.* 2018] Lion HIRTH, Jonathan MÜHLENPFORDT, and Marisa BULKELEY. “The ENTSO-E Transparency Platform – A review of Europe’s most ambitious electricity data platform”. *Applied Energy* 225 (Sept. 2018), pp. 1054–1067. ISSN: 03062619. DOI: [10.1016/j.apenergy.2018.04.048](https://doi.org/10.1016/j.apenergy.2018.04.048) (cit. on pp. 14, 24).
- [HÖLBLING *et al.* 2025] Stefanie HÖLBLING, Gottfried KIRCHENGAST, Christian BRIESE, and Hans THÜMINGER. “Energy use and carbon emissions in high-performance computing: A case study for universities and reduction strategies”. *Cleaner Environmental Systems* 19 (2025). DOI: [10.1016/j.cesys.2025.100332](https://doi.org/10.1016/j.cesys.2025.100332) (cit. on p. 10).

REFERENCES

- [HOUESSIONON *et al.* 2021] M. G. Karel HOUESSIONON *et al.* “Environmental Heavy Metal Contamination from Electronic Waste (E-Waste) Recycling Activities Worldwide: A Systematic Review from 2005 to 2017”. *International Journal of Environmental Research and Public Health* 18.7 (Mar. 2021), p. 3517. ISSN: 1660-4601. DOI: [10.3390/ijerph18073517](https://doi.org/10.3390/ijerph18073517) (cit. on p. 10).
- [INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE 2015] INTERGOVERNMENTAL PANEL ON CLIMATE CHANGE. *Climate Change 2014: Mitigation of Climate Change: Working Group III Contribution to the IPCC Fifth Assessment Report*. 1st ed. Cambridge University Press, Jan. 2015. ISBN: 978-1-107-05821-7 978-1-107-41541-6 978-1-107-65481-5. DOI: [10.1017/CBO9781107415416](https://doi.org/10.1017/CBO9781107415416) (cit. on p. 14).
- [JIANG *et al.* 2025] Yankai JIANG, Raghavendra KANAKAGIRI, Rohan BASU ROY, and Devesh TIWARI. “ThirstyFLOPS: Water Footprint Modeling and Analysis Toward Sustainable HPC Systems”. In: *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*. St. Louis MO USA: ACM, Nov. 2025, pp. 855–869. ISBN: 979-8-4007-1466-5. DOI: [10.1145/3712285.3759804](https://doi.org/10.1145/3712285.3759804) (cit. on pp. 10, 26).
- [KAHIL *et al.* 2025] Hussain KAHIL, Shiva SHARMA, Petri VÄLISUO, and Mohammed ELMUSRATI. “Reinforcement learning for data center energy efficiency optimization: A systematic literature review and research roadmap”. *Applied Energy* 389 (July 2025), p. 125734. ISSN: 03062619. DOI: [10.1016/j.apenergy.2025.125734](https://doi.org/10.1016/j.apenergy.2025.125734) (cit. on pp. 5, 6).
- [KAMATAR *et al.* 2025] Alok KAMATAR *et al.* “Core hours and carbon credits: Incentivizing sustainability in HPC”. In: *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*. Sc ’25. New York, NY, USA: Association for Computing Machinery, 2025, pp. 870–887. ISBN: 979-8-4007-1466-5. DOI: [10.1145/3712285.3759858](https://doi.org/10.1145/3712285.3759858) (cit. on p. 25).
- [KARANIKOLAOU and BEKAKOS 2022] E. M. KARANIKOLAOU and M. P. BEKAKOS. “Action: A New Metric for Evaluating the Energy Efficiency on High Performance Computing Platforms (ranked on Green500 List)”. *WSEAS TRANSACTIONS ON COMPUTERS* 21 (Jan. 2022), pp. 23–30. ISSN: 2224-2872, 1109-2750. DOI: [10.37394/23205.2022.21.4](https://doi.org/10.37394/23205.2022.21.4) (cit. on p. 26).
- [KITCHENHAM and CHARTERS 2007] Barbara KITCHENHAM and Stuart CHARTERS. “Guidelines for performing systematic literature reviews in software engineering”. 2 (Jan. 2007) (cit. on p. 6).
- [KOCOT *et al.* 2023] Bartłomiej KOCOT, Paweł CZARNUL, and Jerzy PROFICZ. “Energy-Aware Scheduling for High-Performance Computing Systems: A Survey”. *Energies* 16.2 (Jan. 2023), p. 890. ISSN: 1996-1073. DOI: [10.3390/en16020890](https://doi.org/10.3390/en16020890) (cit. on pp. 1, 5, 6).

- [KOOMEY *et al.* 2011] Jonathan KOOMEY, Stephen BERARD, Marla SANCHEZ, and Henry WONG. “Implications of Historical Trends in the Electrical Efficiency of Computing”. *IEEE Annals of the History of Computing* 33.3 (Mar. 2011), pp. 46–54. ISSN: 1058-6180, 1934-1547. DOI: [10.1109/MAHC.2010.28](https://doi.org/10.1109/MAHC.2010.28) (cit. on p. 1).
- [KOZLOV and STAMATAKIS 2024] Oleksiy M. KOZLOV and Alexandros STAMATAKIS. “EcoFreq: Compute with cheaper, cleaner energy via carbon-aware power scaling”. In: *ISC High Performance 2024 Research Paper Proceedings (39th International Conference)*. May 2024, pp. 1–12. DOI: [10.23919/isc.2024.10528928](https://doi.org/10.23919/isc.2024.10528928) (cit. on p. 24).
- [KUMAR and PINDORIYA 2026] Amit KUMAR and Naran M. PINDORIYA. “Toward sustainable data center operation: A review on existing infrastructures, integrated smart energy management frameworks, and future perspectives”. *Renewable and Sustainable Energy Reviews* 230 (Apr. 2026), p. 116664. ISSN: 13640321. DOI: [10.1016/j.rser.2025.116664](https://doi.org/10.1016/j.rser.2025.116664) (cit. on pp. 5, 6).
- [LANNELONGUE *et al.* 2021] Loïc LANNELONGUE, Jason GREALEY, and Michael INOUE. “Green Algorithms: Quantifying the Carbon Footprint of Computation”. *Advanced Science* 8.12 (June 2021), p. 2100707. ISSN: 2198-3844, 2198-3844. DOI: [10.1002/advsc.202100707](https://doi.org/10.1002/advsc.202100707) (cit. on p. 11).
- [LI *et al.* 2023] Baolin LI *et al.* “Toward Sustainable HPC: Carbon Footprint Estimation and Environmental Implications of HPC Systems”. In: *Proceedings of the International Conference for High Performance Computing, Networking, Storage and Analysis*. Denver CO USA: ACM, Nov. 2023, pp. 1–15. ISBN: 979-8-4007-0109-2. DOI: [10.1145/3581784.3607035](https://doi.org/10.1145/3581784.3607035) (cit. on pp. 1, 25, 26).
- [LYNAR *et al.* 2010] Timothy M. LYNAR, SIMON, Ric D. HERBERT, and William J. CHIVERS. “Clustering obsolete computers to reduce e-waste”. 1.1 (Jan. 2010), pp. 1–10. ISSN: 1941-868x. DOI: [10.4018/jissc.2010092901](https://doi.org/10.4018/jissc.2010092901) (cit. on p. 11).
- [MACKNICK *et al.* 2012] J MACKNICK, R NEWMARK, G HEATH, and K C HALLETT. “Operational water consumption and withdrawal factors for electricity generating technologies: a review of existing literature”. *Environmental Research Letters* 7.4 (Dec. 2012), p. 045802. ISSN: 1748-9326. DOI: [10.1088/1748-9326/7/4/045802](https://doi.org/10.1088/1748-9326/7/4/045802) (cit. on p. 14).
- [MU’ALEM and D. FEITELSON 2001] A.W. MU’ALEM and D.G. FEITELSON. “Utilization, predictability, workloads, and user runtime estimates in scheduling the IBM SP2 with backfilling”. *IEEE Transactions on Parallel and Distributed Systems* 12.6 (June 2001), pp. 529–543. ISSN: 10459219. DOI: [10.1109/71.932708](https://doi.org/10.1109/71.932708) (cit. on p. 13).
- [MYTTON 2021] David MYTTON. “Data centre water consumption”. *npj Clean Water* 4.1 (Feb. 2021), p. 11. ISSN: 2059-7037. DOI: [10.1038/s41545-021-00101-w](https://doi.org/10.1038/s41545-021-00101-w) (cit. on p. 1).

REFERENCES

- [NAGARATHINAM *et al.* 2024] Aishwarya NAGARATHINAM, Aarth ChELLASAMY, Kingston ANTONYSAMY, Vellaiyan SARAVANAN, and Muthu GOPALAKRISHNAN. “Green Data Centers: A Review of Current Trends and Practices”. In: *Technology-Driven Business Innovation*. Ed. by Rim EL KHOURY. Vol. 223. Cham: Springer Nature Switzerland, 2024, pp. 251–264. ISBN: 978-3-031-51996-3 978-3-031-51997-0. DOI: [10.1007/978-3-031-51997-0_21](https://doi.org/10.1007/978-3-031-51997-0_21) (cit. on pp. 5, 6).
- [NIEWENHUIS *et al.* 2024] Dante NIEWENHUIS, Sacheendra TALLURI, Alexandru IOSUP, and Tiziano DE MATTEIS. “FootPrinter: Quantifying Data Center Carbon Footprint”. In: *Companion of the 15th ACM/SPEC International Conference on Performance Engineering*. London United Kingdom: ACM, May 2024, pp. 189–195. ISBN: 979-8-4007-0445-1. DOI: [10.1145/3629527.3651419](https://doi.org/10.1145/3629527.3651419) (cit. on p. 25).
- [O’BRIEN *et al.* 2018] Kenneth O’BRIEN, Ilia PIETRI, Ravi REDDY, Alexey LASTOVETSKY, and Rizos SAKELLARIOU. “A Survey of Power and Energy Predictive Models in HPC Systems and Applications”. *ACM Computing Surveys* 50.3 (May 2018), pp. 1–38. ISSN: 0360-0300, 1557-7341. DOI: [10.1145/3078811](https://doi.org/10.1145/3078811) (cit. on pp. 5, 6).
- [PAIPURI 2025] Mahendra PAIPURI. “CEEMS: a resource manager agnostic energy and emissions monitoring stack”. In: *Proceedings of the SC ’24 Workshops of the International Conference on High Performance Computing, Network, Storage, and Analysis*. Sc-w ’24. Atlanta, GA, USA: IEEE Press, 2025, pp. 1862–1866. ISBN: 979-8-3503-5554-3. DOI: [10.1109/scw63240.2024.00233](https://doi.org/10.1109/scw63240.2024.00233) (cit. on p. 25).
- [PENG *et al.* 2023] Hua-Rong PENG, Yue-Jun ZHANG, and Jing-Yue LIU. “The energy rebound effect of digital development: Evidence from 285 cities in China”. *Energy* 270 (May 2023), p. 126837. ISSN: 03605442. DOI: [10.1016/j.energy.2023.126837](https://doi.org/10.1016/j.energy.2023.126837) (cit. on p. 11).
- [PETERSEN *et al.* 2008] Kai PETERSEN, Robert FELDT, Shahid MUJTABA, and Michael MATTSSON. “Systematic Mapping Studies in Software Engineering”. In: *12th International Conference on Evaluation and Assessment in Software Engineering (EASE)*. June 2008. DOI: [10.14236/ewic/EASE2008.8](https://doi.org/10.14236/ewic/EASE2008.8) (cit. on pp. 5, 6).
- [PIRSON *et al.* 2023] Thibault PIRSON *et al.* “The Environmental Footprint of IC Production: Review, Analysis, and Lessons From Historical Trends”. *IEEE Transactions on Semiconductor Manufacturing* 36.1 (Feb. 2023), pp. 56–67. ISSN: 0894-6507, 1558-2345. DOI: [10.1109/TSM.2022.3228311](https://doi.org/10.1109/TSM.2022.3228311) (cit. on p. 1).
- [PORTEGIES ZWART 2020] Simon PORTEGIES ZWART. “The ecological impact of high-performance computing in astrophysics”. *Nature Astronomy* 4.9 (Sept. 2020), pp. 819–822. ISSN: 2397-3366. DOI: [10.1038/s41550-020-1208-y](https://doi.org/10.1038/s41550-020-1208-y) (cit. on pp. 1, 11).

- [RAO and CHIEN 2025] Varsha RAO and Andrew A. CHIEN. “Modeling the Carbon Footprint of HPC: The Top 500 and EasyC”. In: *Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis*. St Louis MO USA: ACM, Nov. 2025, pp. 2032–2040. ISBN: 979-8-4007-1871-7. DOI: [10.1145/3731599.3767567](https://doi.org/10.1145/3731599.3767567) (cit. on pp. 10, 26, 27).
- [REINBIGLER and MARTIN 2025] Marie REINBIGLER and Rosalie MARTIN. “Toward a metal usage effectiveness metric in data centers: a study”. In: *2025 11th International Conference on ICT for Sustainability (ICT4S)*. June 2025, pp. 153–164. DOI: [10.1109/ict4s68164.2025.00024](https://doi.org/10.1109/ict4s68164.2025.00024) (cit. on p. 11).
- [ROSA et al. 2026] Lucas de Sousa ROSA et al. *The Environmental Impacts of High-Performance Computing: A Systematic Mapping Study*. 2026 (cit. on pp. 1, 7, 29, 30).
- [SAFARI et al. 2025] Ashkan SAFARI, Hoda SOROURI, Afshin RAHIMI, and Arman OSHNOEI. “A Systematic Review of Energy Efficiency Metrics for Optimizing Cloud Data Center Operations and Management”. *Electronics* 14.11 (May 2025), p. 2214. ISSN: 2079-9292. DOI: [10.3390/electronics14112214](https://doi.org/10.3390/electronics14112214) (cit. on pp. 5, 6).
- [SANDHU et al. 2025] Sofia SANDHU, Ayoub ZUMEIT, Zhenyu TIAN, Vincenzo VINCIGUERRA, and Ravinder DAHIYA. “Semiconductor manufacturing wastewater challenges and the potential solutions via printed electronics”. *iScience* 28.10 (Oct. 2025), p. 113576. ISSN: 25890042. DOI: [10.1016/j.isci.2025.113576](https://doi.org/10.1016/j.isci.2025.113576) (cit. on pp. 1, 10).
- [SANTARIUS et al. 2022] Tilman SANTARIUS et al. “Digital sufficiency: conceptual considerations for ICTs on a finite planet”. *Annals of Telecommunications* 78.5-6 (May 2022), pp. 277–295. ISSN: 0003-4347, 1958-9395. DOI: [10.1007/s12243-022-00914-x](https://doi.org/10.1007/s12243-022-00914-x) (cit. on p. 26).
- [SARAIVA et al. 2025] Gabriella SARAIVA, Miguel VASCONCELOS, Sarita Mazzini BRUSCHI, Danilo CARASTAN-SANTOS, and Daniel CORDEIRO. *Estimating CO2 Emissions of Distributed Applications and Platforms with SimGrid/Batsim*. 2025. DOI: [10.48550/ARXIV.2508.13693](https://doi.org/10.48550/ARXIV.2508.13693) (cit. on p. 14).
- [SCHERER et al. 2023] Laura SCHERER et al. “Biodiversity Impact Assessment Considering Land Use Intensities and Fragmentation”. *Environmental Science & Technology* 57.48 (Dec. 2023), pp. 19612–19623. ISSN: 0013-936X, 1520-5851. DOI: [10.1021/acs.est.3c04191](https://doi.org/10.1021/acs.est.3c04191) (cit. on p. 11).
- [SCHNEIDER ELECTRIC 2023] SCHNEIDER ELECTRIC. *Demystifying Data Center Scope 3 Carbon*. <https://blog.se.com/datacenter/2023/07/11/demystifying-data-center-scope-3-carbon-with-our-findings/>. 2023 (cit. on p. 11).
- [SHELDON et al. 2015] Heather A. SHELDON et al. “Groundwater cooling of a supercomputer in Perth, Western Australia: hydrogeological simulations and thermal sustainability”. *Hydrogeology Journal* 23.8 (2015), pp. 1831–1849. DOI: [10.1007/s10040-015-1280-z](https://doi.org/10.1007/s10040-015-1280-z) (cit. on p. 10).

REFERENCES

- [SHI *et al.* 2025] Tianyao SHI, Ritbik KUMAR, Inez HUA, and Yi DING. “When servers meet species: a fab-to-grave lens on computing’s biodiversity impact”. *SIGENERGY Energy Inform. Rev.* 5.2 (Aug. 2025), pp. 34–40. DOI: [10.1145/3757892.3757897](https://doi.org/10.1145/3757892.3757897) (cit. on p. 11).
- [SHMUELI and D. G. FEITELSON 2007] E. SHMUELI and D. G. FEITELSON. “Uncovering the Effect of System Performance on User Behavior from Traces of Parallel Systems”. In: *2007 15th International Symposium on Modeling, Analysis, and Simulation of Computer and Telecommunication Systems*. Istanbul, Turkey: IEEE, Oct. 2007, pp. 274–280. ISBN: 978-1-4244-1853-4. DOI: [10.1109/MASCOTS.2007.67](https://doi.org/10.1109/MASCOTS.2007.67) (cit. on p. 25).
- [SHUJA *et al.* 2016] Junaid SHUJA, Abdullah GANI, Shahaboddin SHAMSHIRBAND, Raja Wasim AHMAD, and Kashif BILAL. “Sustainable Cloud Data Centers: A survey of enabling techniques and technologies”. *Renewable and Sustainable Energy Reviews* 62 (Sept. 2016), pp. 195–214. ISSN: 13640321. DOI: [10.1016/j.rser.2016.04.034](https://doi.org/10.1016/j.rser.2016.04.034) (cit. on pp. 5, 6).
- [SICKINGER *et al.* 2018] David E. SICKINGER, Otto D. VAN GEET, Suzanne A. BELMONT, Thomas CARTER, and David MARTINEZ. *Thermosyphon Cooler Hybrid System for Water Savings in an Energy-Efficient HPC Data Center: Results from 24 Months and the Impact on Water Usage Effectiveness*. Tech. rep. NREL/TP–2C00-72196, 1471661. Sept. 2018, NREL/TP–2C00-72196, 1471661. DOI: [10.2172/1471661](https://doi.org/10.2172/1471661) (cit. on p. 10).
- [SILVA *et al.* 2024] C.A. SILVA, R. VILAÇA, A. PEREIRA, and R.J. BESSA. “A review on the decarbonization of high-performance computing centers”. *Renewable and Sustainable Energy Reviews* 189 (Jan. 2024), p. 114019. ISSN: 13640321. DOI: [10.1016/j.rser.2023.114019](https://doi.org/10.1016/j.rser.2023.114019) (cit. on pp. 5, 6).
- [SKOVIRA *et al.* 1996] Joseph SKOVIRA, Waiman CHAN, Honbo ZHOU, and David LIFKA. “The EASY – LoadLeveler API project”. In: *Job Scheduling Strategies for Parallel Processing*. Ed. by Gerhard GOOS, Juris HARTMANIS, J. VAN LEEUWEN, Dror G. FEITELSON, and Larry RUDOLPH. Vol. 1162. Berlin, Heidelberg: Springer Berlin Heidelberg, 1996, pp. 41–47. ISBN: 978-3-540-61864-5 978-3-540-70710-3. DOI: [10.1007/BFb0022286](https://doi.org/10.1007/BFb0022286) (cit. on pp. 2, 13).
- [SOLÓRZANO *et al.* 2024] Ana Luisa Veroneze SOLÓRZANO *et al.* “Toward Sustainable HPC: In-Production Deployment of Incentive-Based Power Efficiency Mechanism on the Fugaku Supercomputer”. In: *SC24: International Conference for High Performance Computing, Networking, Storage and Analysis*. Nov. 2024, pp. 1–16. DOI: [10.1109/SC41406.2024.00030](https://doi.org/10.1109/SC41406.2024.00030) (cit. on p. 25).
- [SUKPRASERT *et al.* 2024] Thanathorn SUKPRASERT, Abel SOUZA, Noman BASHIR, David IRWIN, and Prashant SHENOY. “On the Limitations of Carbon-Aware Temporal and Spatial Workload Shifting in the Cloud”. In: *Proceedings of the Nineteenth European Conference on Computer Systems*. Athens Greece: ACM, Apr. 2024, pp. 924–941. ISBN: 979-8-4007-0437-6. DOI: [10.1145/3627703.3650079](https://doi.org/10.1145/3627703.3650079) (cit. on p. 23).

- [UNITED NATIONS 2022] UNITED NATIONS, ed. *Carbon Neutrality in the UNECE Region: Integrated Life-Cycle Assessment of Electricity Sources*. United Nations Publication. Geneva: United Nations, 2022. ISBN: 978-92-1-001485-4 978-92-1-117298-0 (cit. on p. 14).
- [UWIZEYIMANA and JERGER 2025] Iris UWIZEYIMANA and Natalie Enright JERGER. “Carbon-aware server replacement”. In: *2025 IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)*. May 2025, pp. 1–3. DOI: [10.1109/ispass64960.2025.00046](https://doi.org/10.1109/ispass64960.2025.00046) (cit. on pp. 11, 25).
- [WASSERMANN *et al.* 2024] Christian WASSERMANN *et al.* “Calculating User-Centric Carbon Footprints for HPC”. In: *2024 IEEE International Conference on Cluster Computing Workshops (CLUSTER Workshops)*. Sept. 2024, pp. 26–35. DOI: [10.1109/CLUSTERWorkshops61563.2024.00015](https://doi.org/10.1109/CLUSTERWorkshops61563.2024.00015) (cit. on p. 25).
- [WHITEHEAD *et al.* 2014] Beth WHITEHEAD, Deborah ANDREWS, Amip SHAH, and Graeme MAIDMENT. “Assessing the environmental impact of data centres part 1: Background, energy use and metrics”. *Building and Environment* 82 (Dec. 2014), pp. 151–159. ISSN: 03601323. DOI: [10.1016/j.buildenv.2014.08.021](https://doi.org/10.1016/j.buildenv.2014.08.021) (cit. on p. 6).
- [WHITEHEAD *et al.* 2015] Beth WHITEHEAD, Deborah ANDREWS, Amip SHAH, and Graeme MAIDMENT. “Assessing the environmental impact of data centres part 2: Building environmental assessment methods and life cycle assessment”. *Building and Environment* 93 (Nov. 2015), pp. 395–405. ISSN: 03601323. DOI: [10.1016/j.buildenv.2014.08.015](https://doi.org/10.1016/j.buildenv.2014.08.015) (cit. on p. 6).
- [WIESNER *et al.* 2021] Philipp WIESNER, Ilja BEHNKE, Dominik SCHEINERT, Kordian GONTARSKA, and Lauritz THAMSEN. “Let’s wait awhile: how temporal workload shifting can reduce carbon emissions in the cloud”. In: *Proceedings of the 22nd International Middleware Conference*. Québec city Canada: ACM, Dec. 2021, pp. 260–272. ISBN: 978-1-4503-8534-3. DOI: [10.1145/3464298.3493399](https://doi.org/10.1145/3464298.3493399) (cit. on p. 23).
- [YANG and WANG 2023] Haodong YANG and Gaofeng WANG. “The impact of computing infrastructure on carbon emissions: an empirical study based on china national supercomputing center”. *Environmental Research Communications* 5.9 (2023). DOI: [10.1088/2515-7620/acfa65](https://doi.org/10.1088/2515-7620/acfa65) (cit. on p. 26).
- [YORK *et al.* 2022] Richard YORK, Lazarus ADUA, and Brett CLARK. “The rebound effect and the challenge of moving beyond fossil fuels: A review of empirical and theoretical research”. *WIREs Climate Change* 13.4 (July 2022), e782. ISSN: 1757-7780, 1757-7799. DOI: [10.1002/wcc.782](https://doi.org/10.1002/wcc.782) (cit. on pp. 1, 26).
- [ZHANG *et al.* 2021] Qingxia ZHANG *et al.* “A survey on data center cooling systems: Technology, power consumption modeling and control strategy optimization”. *Journal of Systems Architecture* 119 (Oct. 2021), p. 102253. ISSN: 13837621. DOI: [10.1016/j.sysarc.2021.102253](https://doi.org/10.1016/j.sysarc.2021.102253) (cit. on pp. 5, 6).

REFERENCES

- [ZHAO *et al.* 2025] Paiguang ZHAO, Baowen LOU, Yi LUO, and Zihao PENG. “High-Powered Computing: Energy Consumption and Its Environmental Impact”. *Advances in Engineering Technology Research* 13.1 (Mar. 2025), p. 1337. ISSN: 2790-1688. DOI: [10.56028/aetr.13.1.1337.2025](https://doi.org/10.56028/aetr.13.1.1337.2025) (cit. on p. [11](#)).